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SCHOOL OF ENGINEERING AND TECHNOLOGY ASSIGNMENT COVER SHEET 1

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1.0 Comprehensive Analysis of OSI Protocol Stack

The OSI Protocol Stack

The Open Systems Interconnection (OSI) Model is developed by the International Organization for Standardization (ISO) in the year 1984. The OSI Model is a concept that computer systems use to communicate over a network. It is a framework of rules that computer systems follow to transmit data to each other. The OSI Protocol Stack consists of 7 layers that work in tandem to ensure data is transferred across the net swiftly and securely. Each layer interacts with the layer above it and below it, encapsulating and transmitting data across networks. To improve the understanding of the network layers, visual representations from an article regarding the OSI Model Layers published on lifewire.com are used.

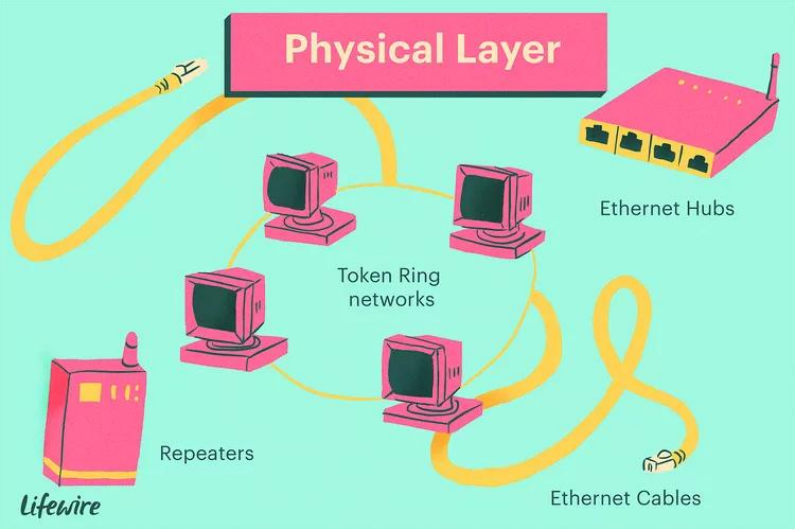


Figure 1.1: Physical Layer

The first layer in the OSI Protocol Stack is the Physical Layer. The Physical Layer is the lowest layer of the OSI Protocol Stack. It consists of the physical cable or connection that is used between the devices. It contains information in bits and transmits individual bits one at a time over a physical medium. When receiving data, this layer will convert the data into 0s and 1s and send it to the Data Link Layer. Some examples of protocols in this layer are DSL, Bluetooth, and Wi-Fi.

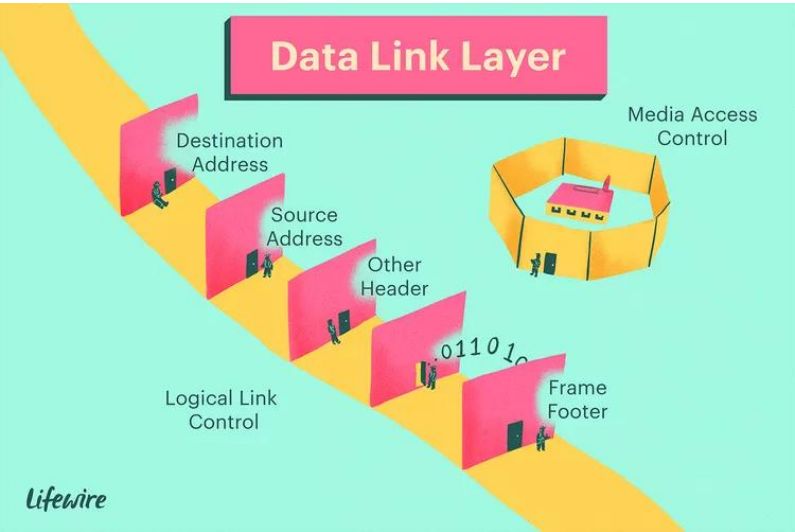


Figure 1.2: Data Link Layer

The second layer of the OSI Protocol Stack is the Data Link Layer. The Data Link Layer is responsible for the node-to-node data transfer and error correction. It divides the bits it receives into easier manageable units called frames. These frames include headers, trailers, and error-checking data. The Data Link Layer also handles physical addressing by transmitting packets to the host using its MAC address. The Data Link Layer also ensures the transmitting of data is error-free. Some examples of protocols in this layer are FDDI, Ethernet, and CDP.

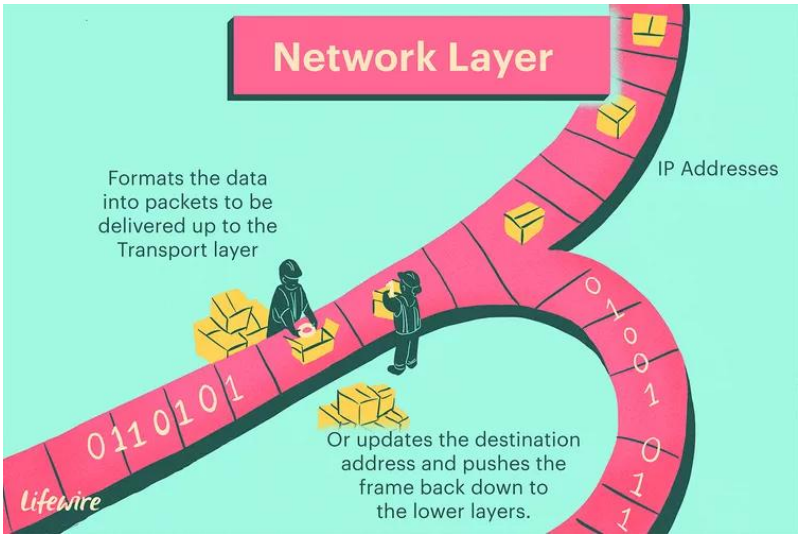


Figure 1.3: Network Layer

The third layer of the OSI Protocol Stack is the Network Layer. The Network Layer provides logical addressing and routing, enabling data transfer between individual networks. It places the sender and receiver's IP addresses in the header of the data. It also ensures that the packets take the best path (the shortest path) through the network from source to destination; this is referred to as packet routing. For example, a router receives a packet from a connected device, it then examines the destination IP that is attached to the header of the data packet, the router then follows its routing table and finds the best path the packet can travel to reach its destination, the router then sends out the packet to the next device and the process is repeated until it reaches its destination. Some examples of protocols in this layer are IPv4, IPv6, and IPX.

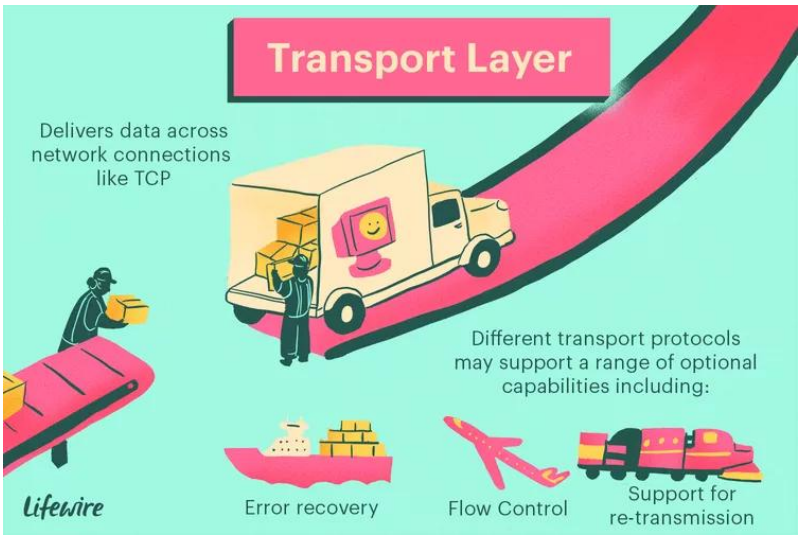


Figure 1.4: Transport Layer

The fourth layer of the OSI Protocol Stack is the Transport Layer. The Transport Layer's main

responsibility is to provide end-to-end communication between applications running on hosts within a network. It divides large data chunks into smaller, more manageable segments for transmission. On the receiving end, it then reassembles the segments back into the original data. This layer is important as it provides reliable data transfer services to upper layers. Some examples of protocols in this layer are User Datagram Protocol (UDP) and Transmission Control Protocol (TCP).

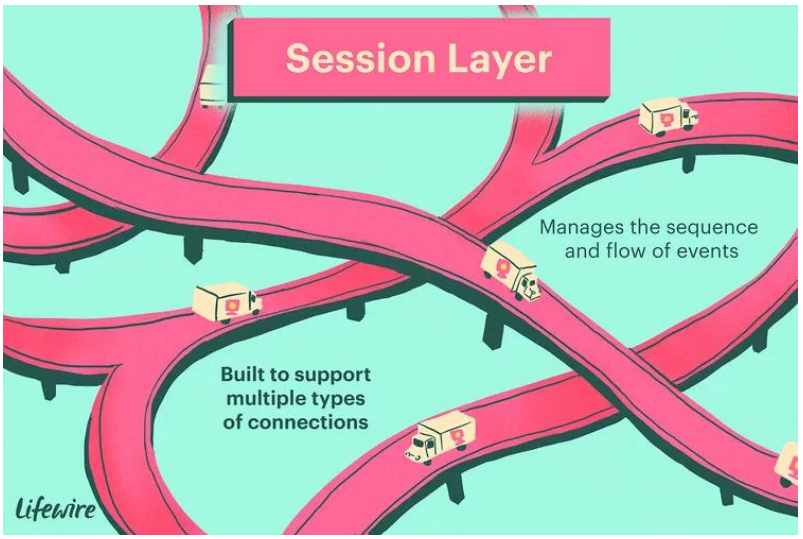


Figure 1.5: Session Layer

The fifth layer of the OSI Protocol Stack is the Session Layer. The Session Layer manages the sessions between applications. It first establishes the connection between the two nodes that are devices. It then is responsible for maintaining this connection and also determining whether communication is full-duplex, half-duplex, or simplex. Full duplex is when both devices can transmit data simultaneously. Half-duplex is when both devices take turns to transmit data. Simplex is when only one device is able to transmit data while the other device is receiving only. The Session Layer is also responsible for terminating the session when necessary. Some examples of protocols in this layer are PPTP and SCP.

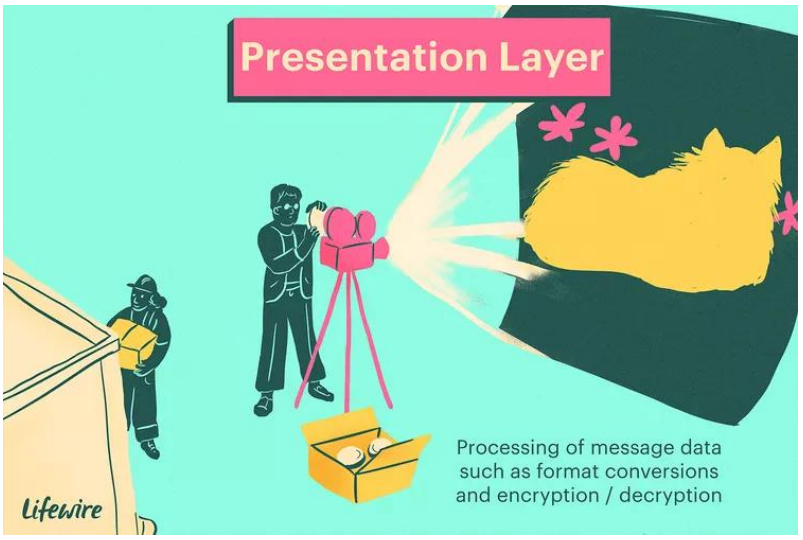


Figure 1.6: Presentation Layer

The sixth layer of the OSI Protocol Stack is the Presentation Layer. The Presentation Layer ensures that the information or data that the application layer of one device transmits is readable by the receiving device. It converts the data into standard and transmittable formats. On the receiving end, it is responsible for translating and formatting the data to the receiving device's format. The Presentation Layer is also capable of compressing and encrypting data. In

order words, this layer formats and encrypts data for the application layer. Some examples of protocols in this layer are JPEG, MPEG, and Secure Sockets Layer (SSL).

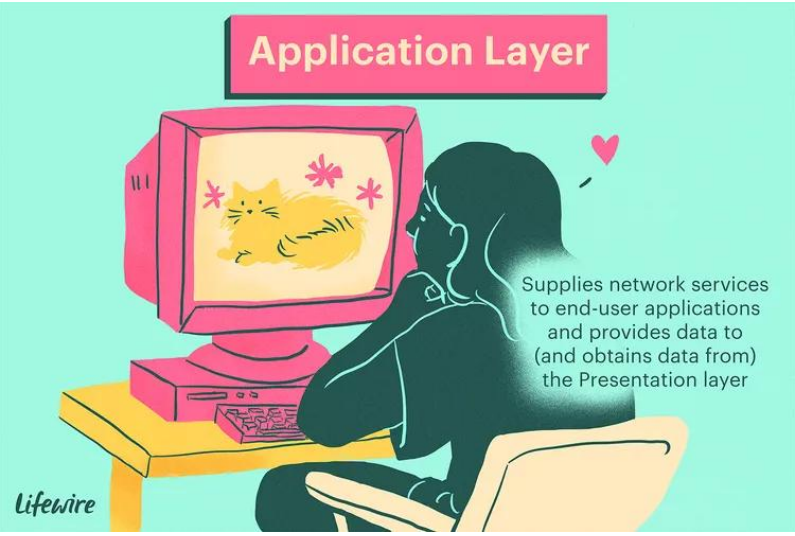


Figure 1.7: Application Layer

The seventh and last layer of the OSI Protocol Stack is the Application Layer. The Application Layer acts as the direct interface that the user of a device interacts with. It is the layer in which users access and utilize network functions. It provides network services to end-user applications as well as allows software to send or receive information and present data to the users. Some examples of protocols in this layer are HTTP, DNS, and SNMP.

In conclusion, the OSI Protocol Stack consists of seven layers: Physical, Data Link, Network, Transport, Session, Presentation, and Application. The Physical Layer transfers bits over wires, The Data Link Layer ensures error-free frame transfer, The Network Layer is responsible for packet routing, The Transport Layer reliably manages end-to-end data flow, The Session Layer is responsible for session management, The Presentation Layer performs data formatting and translation, and lastly the Application Layer is the interface that the user interacts with directly. Conceptually, Layers 1 to 3 focus on data movement and can be referred as the media layers. Layers 5 to 7 focus on data processing and can be referred as the host layers. Lastly, the Transport Layer (Layer 4) acts as a bridge that connects the two groups. All 7 layers work together like a team, each with a specific task to execute.

Advantages of OSI Protocol Stack

1. **Standardization and Interoperability**
   The OSI model provides a systematic framework for network transmission. By defining seven distinct layers with specific functions that ensures the different hardware and software components from various vendors that can interoperate. This standardization facilitates the development of network protocols and devices that cling to a common set of rules. For example, if one device implements the OSI network layer, regardless of the vendor.

2. **Modular Design and Troubleshooting**
   The layered OSI model breaks down the complex network functions into smaller and more manageable components. This helps simplify network design, maintenance, and implementation. When troubleshooting happens the OSI model provides a systematic

approach by examining each layer individually, the network administer can isolate the source of issue that is faced. Example, if a user cannot access the website, the administer would check the physical layer for connection issues, data link layer for addressing and so on.

3. **Facilities Understanding of Complex Systems**
   The OSI model breaks down the very complex process of network communication into smaller pieces of parts. By visualizing the flow of data through the seven layers it helps to clarifies the role of each component and how it interacts. This makes more sense for students, network engineers and other professionals to understand how network functions.

4. **Clear Definitions of Functions**
   The layer of OSI is well defined set of functions. The clear separation makes it easier to understand how network communication works and develop new network protocols. Example, the physical layer deals with the transmission of raw bits while the data link layer handles error detection and correction within local network segment. This clear definition promotes a deeper understanding network operation and allows specialized development of hardware and software foreach layer.

<u>Challenges of OSI Protocol Stack</u>

Despite the multitudes of advantages, the OSI model still poses for challenges and limitations. The following are the challenges of OSI Protocol Stack.

1. **Interoperability Issues**
   The variety of protocols implemented across different vendors could lead to compatibility issues. For example, between different networking devices and software. The standards and services each layer provide are highly theoretical, making it difficult to implement if the appropriate technology is not available.

2. **High Complexity**
   The complexity of the OSI model is rather high resulting in a lack of implementation within modern technologies. Some layers within the OSI model offer very little functionality and do not fulfill the practical needs, such as the presentation layer and session layer. Both layers deal with user interaction and session management respectively. Though they are of practical use, they do not serve much purpose compared to other layers in the model.

3. **Interdependence among the layers**
   Due to the correlation of every 7 layers of the OSI model, each layer cannot work in parallel and must wait for the data from the previous layer before executing its services. This restricts its practical implementation and makes it rather time consuming. This setback makes it difficult to implement as it is not very efficient in terms of operationality, functionality and practicality.

4. **Dominance of TCP/IP Model**
   The TCP/IP model has been widely adopted in modern networking, especially since it was introduced around the time when the Internet was launched. When the OSI model was established, many network professionals have already been implementing the

TCP/IP model within their networking protocols.

Real-World Examples of OSI Protocol Stack

1. **Online Banking Transactions**

   The OSI model is crucial to the safe and effective operation of online banking. The Application Layer (Layer 7) enables user engagement when they log into a banking website or app. To check balances, transfer money and carry out other tasks, the banking application or website interacts with servers. By employing TLS (SSL) encryption, the Transport Layer (Layer 4) makes sure the data is transferred safely and keeps hackers from intercepting private financial data. Data routing between users and bank servers is handled by the Networking Layer (Layer 3). Over Ethernet, Wi-Fi, or mobile networks, the Physical and Data Link Layers (Layers 1 & 2) manage the actual data transfer.

2. **Voice over IP (VoIP) Communication**

   For seamless voice and video communication, VoIP technology is utilized in programs like Zoom, Microsoft Teams, and Skype, which depends on the OSI model. While the Presentation Layer (Layer 6) uses algorithms such as G.711 and Opus, to compress and encrypt audio and video data, the Application Layer (Layer 7) allows users to make calls. Using SIP (Session Initiation Protocol), the Session Layer (Layer 5) makes sure calls are established and maintained correctly. Data packets are transmitted by the Transport Layer (Layer 4) using TCP or UDP. Voice data is routed over the internet by the Network Layer (Layer 3) using IP addresses. Lastly, the Physical and Data Link Layers (Layers 1 & 2) control how data is physically sent via networks such as Ethernet, Wi-Fi and LTE/5G.

3. **Cloud Computing Services**

   The OSI model is used by cloud computing platforms such as Microsoft Azure, Google Cloud and Amazon Web Services (AWS) to deliver scalable and secure services. Users can communicate with cloud-based apps like Google Drive and OneDrive thanks to the Application Layer (Layer 7). To ensure security, the Presentation Layer (Layer 6) encrypts data using the SSL/TLS protocols. To guarantee seamless access to cloud resources, the Session Layer (Layer 5) controls active user sessions. Data is efficiently sent by the Transport Layer (Layer 4) using either QUIC or TCP. In cloud data centers, the Network Layer (Layer 3) makes ensuring that BGP routing and IP addressing are done correctly. Software-Defined Networking (SDN), Ethernet and fiber optic data transmission are managed by the Physical and Data Link Layers (Layers 1 & 2).

2.0 Design and Implementation of a Network

PART A: DESIGN

i. List of Required Equipment

The following is the list of equipment that will be utilized in the network design and the reason of choosing the respective model of the equipment.

No.	Equipment	Model	Reason of Choosing
Intermediary Network Devices			
1.	Router	2911	The modularity, reliable routing with integrated security, is suitable for connecting labs and external networks efficiently. The usage of router is essential to ensure connectivity between the HUMAC lab and the SET labs.
2.	Switches	2960	Cisco 2960 switches were chosen for the labs to provide sufficient port density and Layer 2 functionality. They function as a switch that are connected to the hosts between each lab.
3.	Wireless Access Points	3702i	They will provide wireless connectivity for the wireless devices to the wired network. They allow wireless devices within the AWL and IOT labs to connect to the network.
4.	Power Distributions Unit (PDU)		PDU will provide reliable power to the networking equipment rack
Network Media			
5.	Ethernet Cables	Copper Straight Through	Copper straight-through cables enable connectivity between hosts and network devices. They connect the PCs to switches and to connect the switches to the router.
6.	Serial DTE	Standard Serial DTE	Serial DTEs send or receive data in serial communication, functioning as the end-point device.
End Devices			
7.	PC	Standard PC	PCs will serve purposes of general computing and software development.
8.	Laptop	Standard Laptop	Laptops provide portable computing for productivity, communication, entertainment, and education.
9.	Printer	Standard Printer	Printers create physical copies of digital documents and images onto paper or other media.
10.	Smartphone	Standard Smartphone	A pocket-sized computer with communication apps and internet access
11.	Motion Detector	Standard Motion Detector	It senses movement, triggers an alert or action.

ii. Estimated Cost of Equipment

The project detailed in the tables outlines the establishment of several networking labs at Sunway University.

The following is the **estimated equipment cost** utilized in this project. The budgeting for the equipment cost is categorized according to the 4 respective labs to ensure a more systematic financial plan. Our network infrastructure relies heavily on Cisco equipment, specifically the 2911 router and 2960 switch, which we procure through FS.com (https://www.fs.com/sg) Singapore, a global enterprise-level IT network solution provider. These devices are essential for establishing and maintaining a stable and efficient network environment. FS.com's reliable supply chain ensures that we receive genuine Cisco products, enabling us to leverage their advanced features and robust performance. This allows us to confidently manage network operations, knowing that the core components are dependable and well-supported.

Estimated Equipment Cost

3rd Floor, Sunway University - HUMAC Lab					
No.	Equipment	Model	Quantity	Estimated Cost per Unit (RM)	Total Estimated Cost (RM)
1	Switch	2960	1	2,000	2,000
2	PC	Standard PC	9	1,199	10,971
3	Printer	Standard Printer	3	500	1,500
4	Ethernet Cable	Copper Straight Through	12	51	612
				Subtotal	**15,083**

SET - Communications Lab (CL)					
No.	Equipment	Model	Quantity	Estimated Cost per Unit (RM)	Total Estimated Cost (RM)
1	Switch	2960	1	2,000	2,000
2	PC	Standard PC	5	1,199	5,995
3	Ethernet Cables	Copper Straight Through	5	51	255
				Subtotal	**8,250**

SET - Advanced Wireless Lab (AWL)					
No.	Equipment	Model	Quantity	Estimated Cost per Unit (RM)	Total Estimated Cost (RM)
1	Wireless Access Point	PT-AC	1	1,700	1,700
2	Switch	2960	1	2,000	2,000
3	PC	Standard PC	2	1,199	2,398
4	Laptop	Standard Laptop	5	1,500	7,500
5	Ethernet Cable	Copper Straight Through	3	51	153

				Subtotal	**13,751**

SET - IoT Lab (IOT)					
No.	Equipment	Model	Quantity	Estimated Cost per Unit (RM)	Total Estimated Cost (RM)
1	Wireless Access Point	PT-AC	1	1,700	1,700
2	Switch	2960	1	2,000	2,000
3	PC	Standard PC	7	1,199	8,393
4	Laptop	Standard Laptop	1	1,500	1,500
5	Smartphone	Standard Smartphone	1	500	500
6	Motion Detector	Standard Motion Detector	1	800	800
7	Ethernet Cable	Copper Straight Through	10	51	510
				Subtotal	**15,403**

WAN Link (3rd Floor, Sunway University Router to SET Building Router Connection)					
No.	Equipment	Model	Quantity	Estimated Cost per Unit (RM)	Total Estimated Cost (RM)
1	Router	2911	2	971	1,942
2	Ethernet Cable	Copper Straight Through	4	51	204
3	Serial DTE	Standard	1	80	80
				Subtotal	**2,226**

The total cost for equipment in the HUMAC Lab, Communication Lab, Advanced Wireless Lab, IoT Lab, and WAN Link are RM15083, RM8250, RM13751, RM15403 and RM2226, respectively. The sum of the equipment cost across these 5 connections are RM54713.

Total Equipment Cost		
Lab	Lab Name	Total Estimated Cost (RM)
1	HUMAC Lab	15,083
2	Communications Lab	8,250
4	Advanced Wireless Lab	13,751
3	IoT Lab	15,403
5	WAN Link	2,226
	Total Equipment Cost (RM)	**54,713**

Notably, a significant portion of the networking equipment, specifically the switches (2960) and routers (2911), are products from Cisco Systems, a leading provider of networking solutions. This strategic choice provides students with invaluable hands-on experience using industry-standard hardware, a crucial asset for their future careers.

In addition to the equipment costs, **labor expenses** constitute a substantial portion of the project budget. These costs encompass the installation, configuration, testing, and project management of the network infrastructure. Installation costs refer to costs associated with hiring personnel to perform installation tasks, specifying cable installation, network configuration and testing and verification. Project management costs refer to costs associated with planning, organizing, and managing the project as well as execution of the project. The work labor for configuring, deploying, and testing network devices is performed by technicians to guarantee seamless lab connectivity.

Estimated Labor Cost

Total Labor Cost					
No.	Cost Item	Total Units	Estimated Cost per Unit / Installation (RM)	Estimated Total Cost (RM)	Remarks
1	Cable Installation	35 cable drops	80	2,800	Labor to run & terminate 35 cable drops
2	Network Configuration	41 devices	50	2,050	Labor to configure all devices
3	Testing and Verification	8 devices	60	480	Labor to test and verify network devices
4	Device Installation	41 devices	100	4,100	Labor for physical installation of devices
			Sub Total Cost (RM)	**9,430**	
No.	Cost Item	Number of Workers	Estimated Cost/ Hourly Rate	Estimated Total Cost (RM)	Remarks
1	Project Management	1 project manager	15% of subtotal labor cost	1,415	15% of subtotal labor cost of RM 9,430 = 1414.5 ≈ RM1,415
			Total Labor Cost (RM)	**10,845**	

The total labor cost, which includes the installation cost of RM 9,430 and the project management cost of RM1,415, amounts to RM 10,845.

Grand Total Cost		
No	Cost Item	Estimated Total Cost (RM)
1	Total Equipment Cost	54,713
2	Total Labor Cost	10,845
	Grand Total Cost	**65,558**

Consequently, the grand total for the entire project, combining both equipment and labor costs, is RM65,558. It is important to remember that these figures are specific to the networking lab setup; general electrical service costs in Kuala Lumpur can vary significantly depending on factors such as the type of service, job complexity, and the technician's experience. The investment in these Cisco-equipped networking labs will yield significant benefits for Sunway University students. By utilizing Cisco equipment, students gain practical experience with technologies widely deployed in the industry, enhancing their employability.

iii. Physical Topology

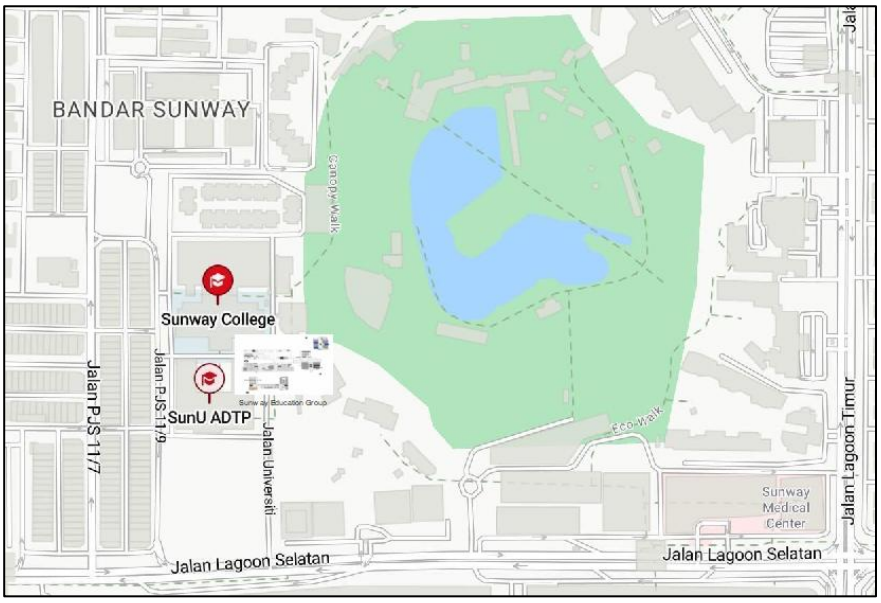


Figure 2.1: Sunway City Map

Upon accessing the Cisco Packet Tracer of this network design, the map of Sunway City, Subang Jaya, Selangor, Malaysia, as shown in ***Figure 2.1***, is displayed. When Sunway Education Group is clicked, it opens up the layout within the campus, consisting of the main buildings we are tasked to connect, SET building and Sunway University 3rd Floor.

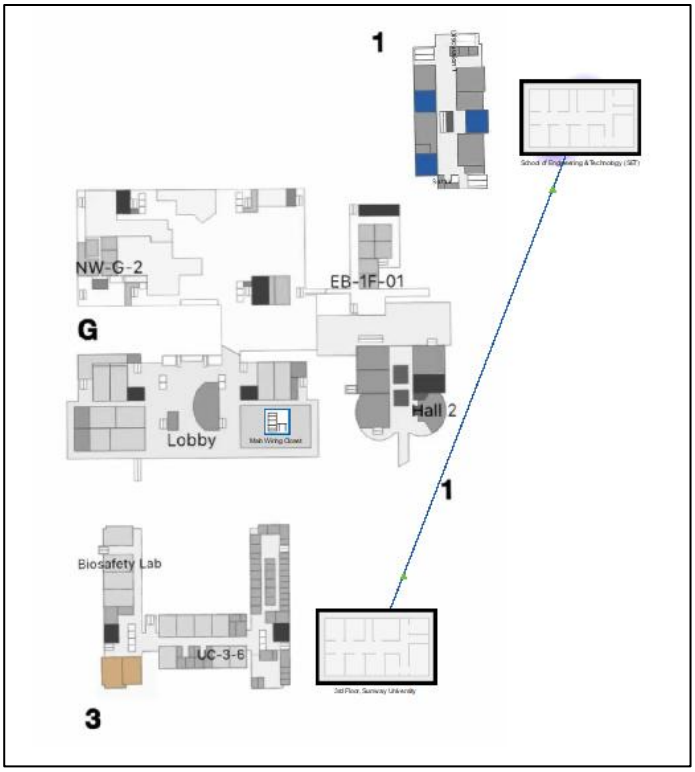


Figure 2.2: Sunway Campus Layout

Figure 2.2 shows the layout of Sunway campus which include Sunway University and the connection with the SET building. There are 2 rooms that can be clicked which are the 3rd Floor in Sunway University and the SET building.

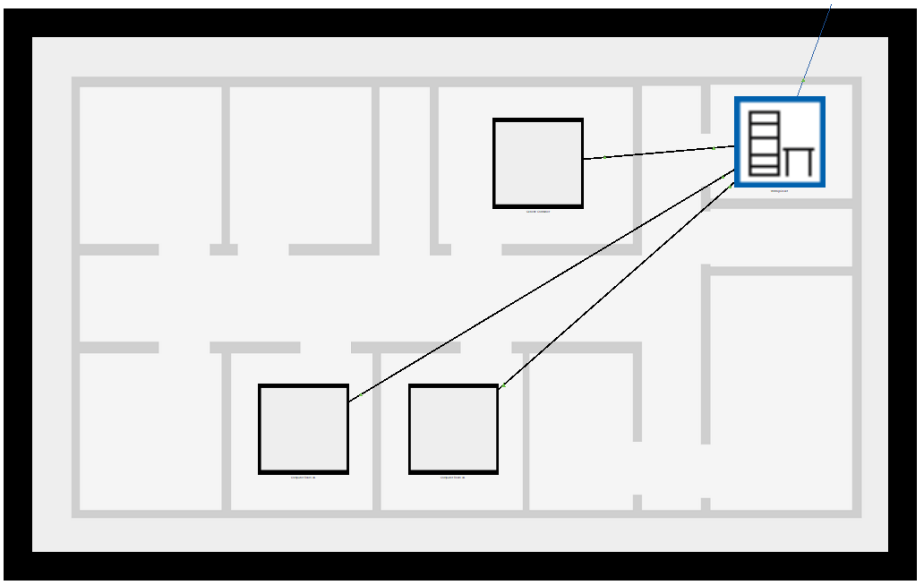


Figure 2.3: 3rd Floor, Sunway University Layout

Figure 2.3 shows the layout of the 3rd Floor in Sunway University. It consists of 4 rooms, Computer Room 3A, Computer Room 3B, Printer Room, and the Main Wiring Closet. This layout also consists of the HUMAC Lab subnet.

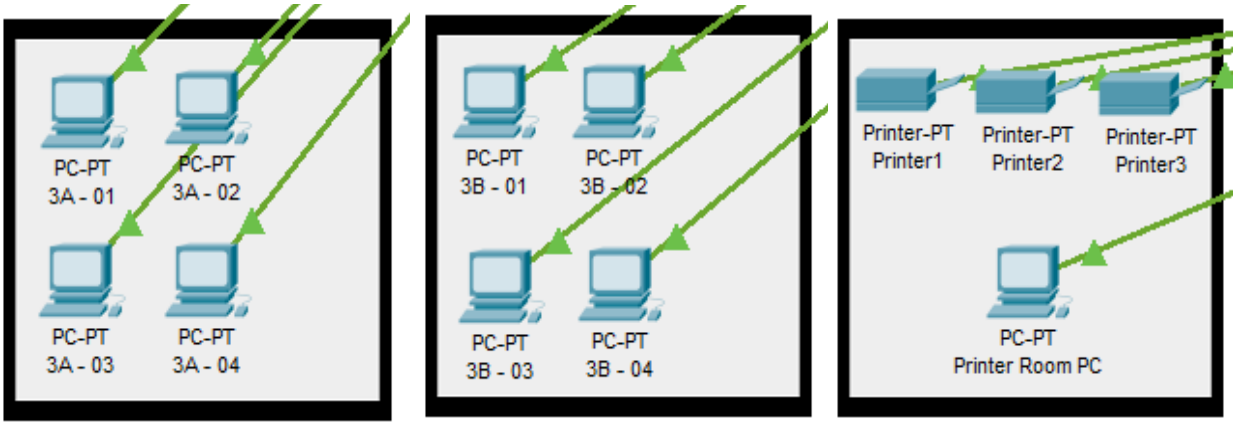


Figure 2.4, 2.5, 2.6: HUMAC Lab Hosts

Figures 2.4, 2.5 and ***2.6*** show the main rooms where the hosts are located. There are 9 PCs and 3 printers in total. These devices make up the hosts of the HUMAC Lab.

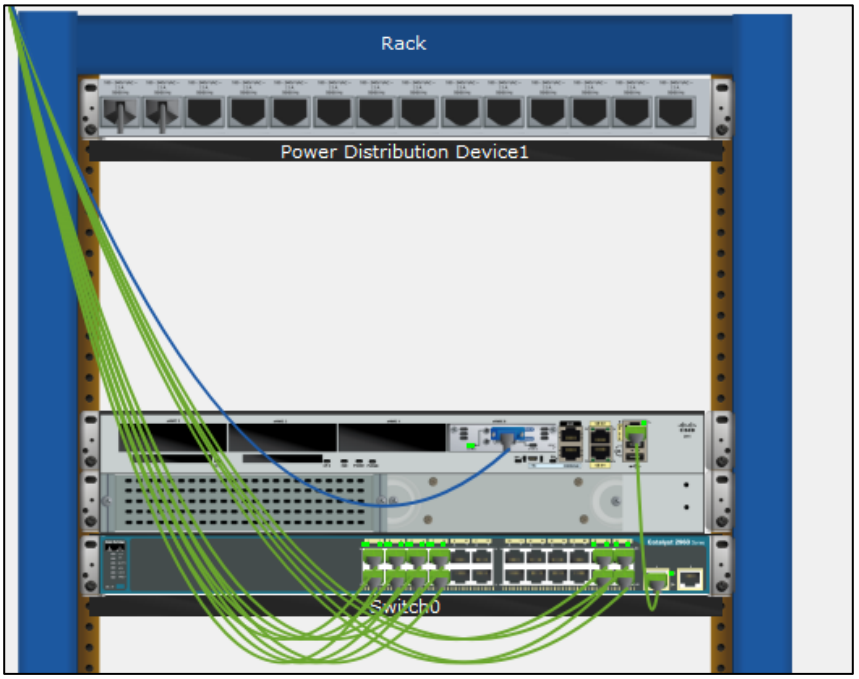


Figure 2.7: HUMAC Lab Main Wiring Closet

Figure 2.7 shows the inside of the Main Wiring Closet in the HUMAC Lab. The Rack in the closet holds the Power Distribution Device, The HUMAC Router, and The HUMAC Switch. The green cables are copper straight through cables that are used to connect to the end devices and to connect the Router to the Switch. The blue cable represents the Serial DCE cable that is connected to the Router in the SET Building, creating the WAN subnet.

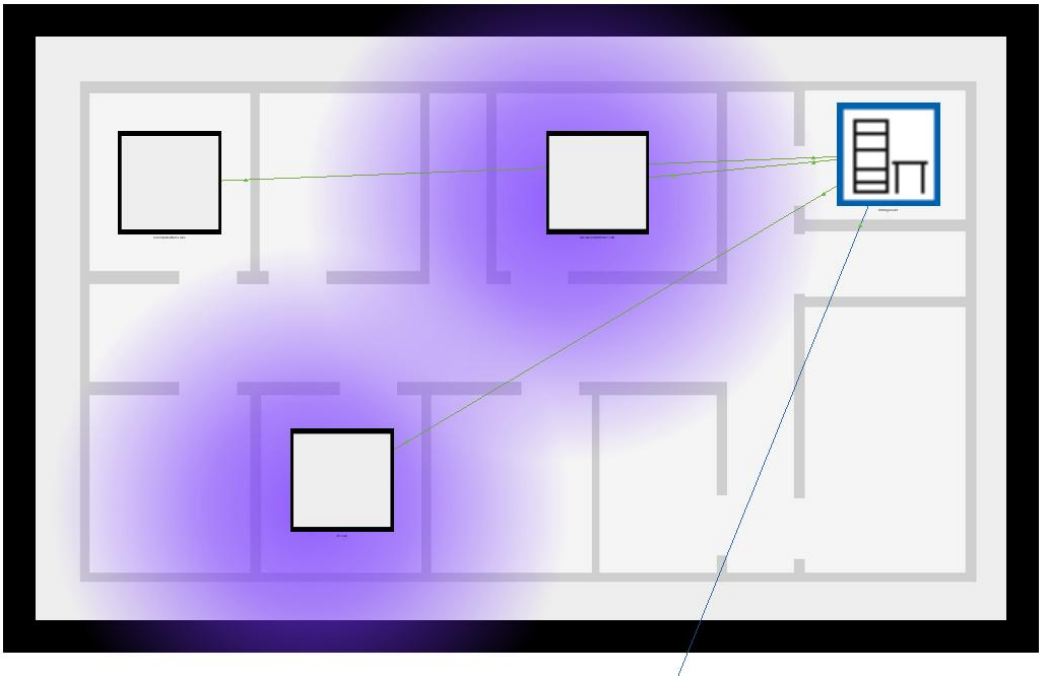


Figure 2.8: SET Building Layout

Figure 2.8 shows the SET Building subnet. It consists of 4 rooms, Communications Lab (CL), Advanced Wireless Lab (AWL), IoT Lab (IOT), and the Main Wiring Closet.

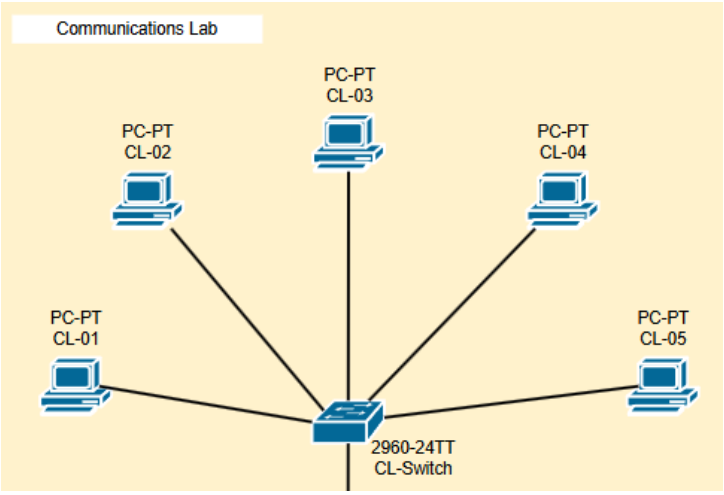


Figure 2.9: Communications Lab Hosts

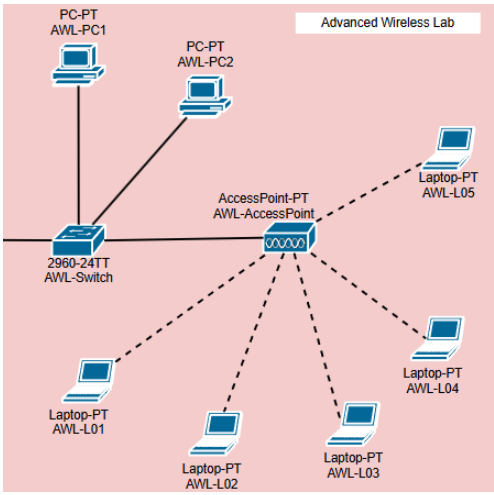


Figure 2.10: Advanced Wireless Lab Hosts

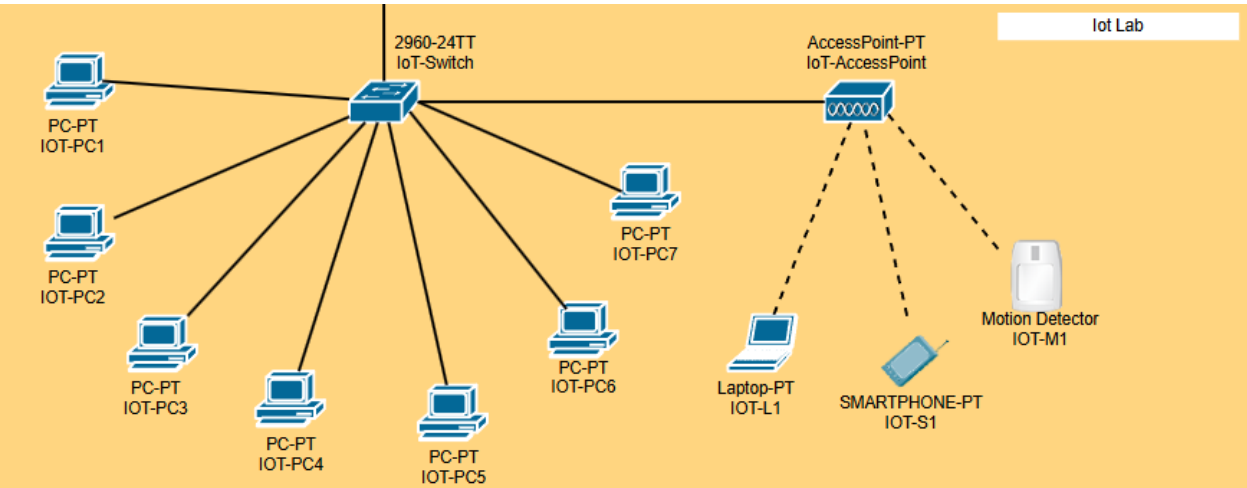


Figure 2.11: IoT Lab Hosts

The figures above show the inside of the three labs that are located in the SET building. The Communications Lab *(**Figure 2.9**)* consists of 5 PCs with its own switch. The AWL Lab *(**Figure 2.10**)* consists of 2 PCs, an access point, 5 laptops for students to connect to the access point and its own switch. The IoT Lab *(**Figure 2.11**)* consists of 7 PC's, 1 access point that is connected to 2 IoT devices and a laptop that students can use to test the IoT devices.

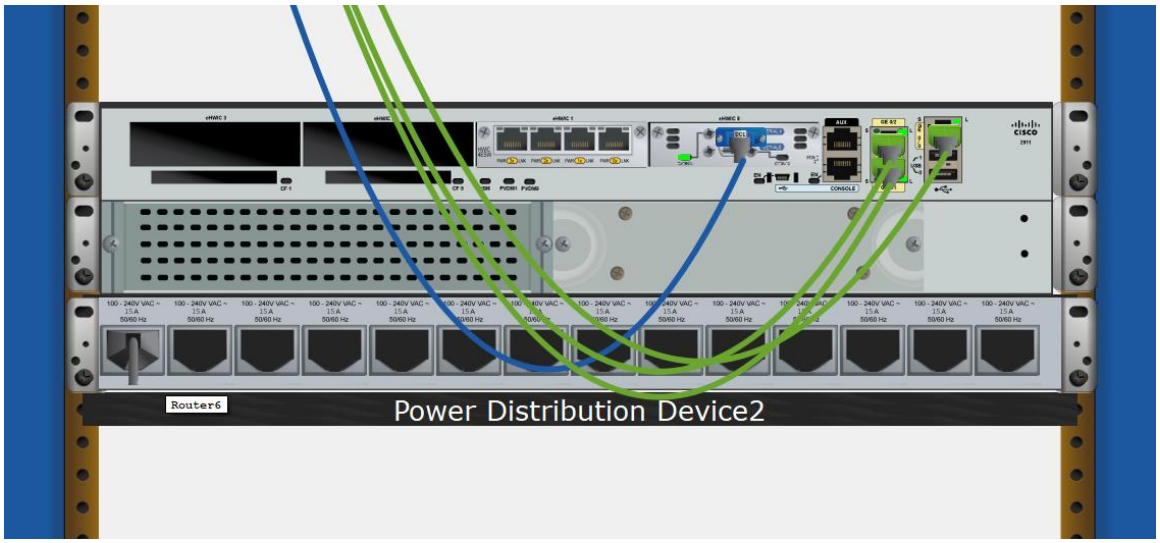


Figure 2.12: SET Building Wiring Closet

Figure 2.12 shows the Wiring Closet that houses the central router in the SET building. This router is connected to the 3 switches in the Communications Lab, AWL Lab and IoT Lab via the 3 green cables, which are copper straight through cables. The blue cable, a serial DCE cable, is used to create the WAN connection between the router in the SET building and the router located at the 3rd floor of the University Building. There is also a Power Distribution Device that provides power to the switch placed below.

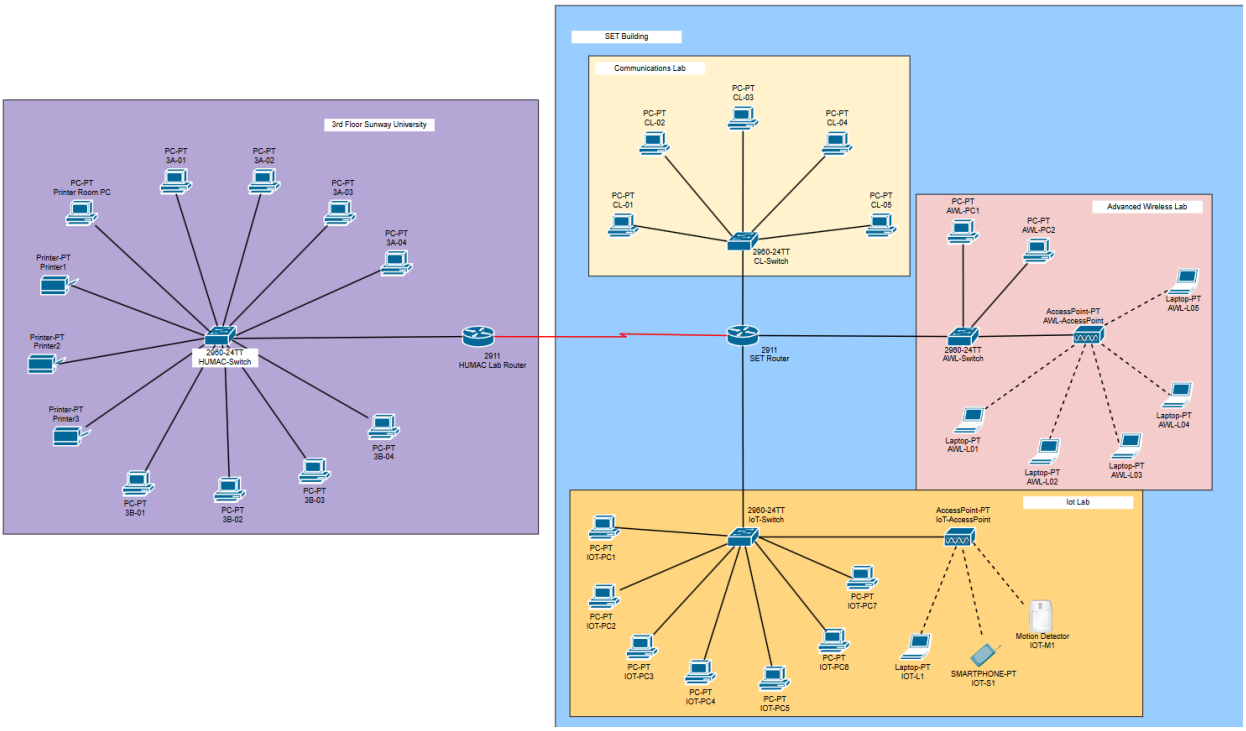


Figure 2.13: Physical Topology

Figure 2.13 shows the physical topology of the network. The network is built within the Sunway campus which includes the 3rd Floor of Sunway University *(purple coloured)* and the new SET Building *(blue coloured)*.

The following displays the physical connections between devices, specifying the source and destination ports of each device.

Physical Connections

	SET Router	
Device	Source Port	Router Port
HUMAC Router	Serial0/0/0	Serial0/0/0
CL - Switch	GigabitEthernet0/1	GigabitEthernet0/2
AWL - Switch	GigabitEthernet0/1	GigabitEthernet0/1
IOT - Switch	GigabitEthernet0/1	GigabitEthernet0/0
	HUMAC Router	
Device	Source Port	Router Port

SET Router	Serial0/0/0	Serial0/0/0
HUMAC - Switch	GigabitEthernet0/1	GigabitEthernet0/0
	HUMAC - Switch	
Device	Source Port	Switch Port
HUMAC Router	Serial0/0/0	Serial0/0/0
3A - 01	FastEthernet0	FastEthernet0/1
3A - 02	FastEthernet0	FastEthernet0/2
3A - 03	FastEthernet0	FastEthernet0/3
3A - 04	FastEthernet0	FastEthernet0/4
3B - 01	FastEthernet0	FastEthernet0/5
3B - 02	FastEthernet0	FastEthernet0/6
3B - 03	FastEthernet0	FastEthernet0/7
3B - 04	FastEthernet0	FastEthernet0/8
Printer Room PC	FastEthernet0	FastEthernet0/21
Printer 1	FastEthernet0	FastEthernet0/22
Printer 2	FastEthernet0	FastEthernet0/23
Printer 3	FastEthernet0	FastEthernet0/24
	CL - Switch	
Device	Source Port	Switch Port
SET Router	GigabitEthernet0/2	GigabitEthernet0/1
CL - 01	FastEthernet0	FastEthernet0/1
CL - 02	FastEthernet0	FastEthernet0/2
CL - 03	FastEthernet0	FastEthernet0/3
CL - 04	FastEthernet0	FastEthernet0/4
CL - 05	FastEthernet0	FastEthernet0/5
	AWL - Switch	
Device	Source Port	Switch Port
SET Router	GigabitEthernet0/1	GigabitEthernet0/1
AWL - PC1	FastEthernet0	FastEthernet0/1
AWL - PC2	FastEthernet0	FastEthernet0/2
AWL – Access Point	GigabitEthernet0/2	Port0
	IOT - Switch	
Device	Source Port	Switch Port
SET Router	GigabitEthernet0/0	GigabitEthernet0/1

IOT – Access Point	GigabitEthernet0/1	Port0
IOT - PC1	FastEthernet0	FastEthernet0/7
IOT - PC2	FastEthernet0	FastEthernet0/6
IOT - PC3	FastEthernet0	FastEthernet0/5
IOT - PC4	FastEthernet0	FastEthernet0/4
IOT - PC5	FastEthernet0	FastEthernet0/3
IOT - PC6	FastEthernet0	FastEthernet0/2
IOT - PC7	FastEthernet0	FastEthernet0/1
	AWL – Access Point	
Device	Source Port	Access Point Port
AWL - Switch	Port0	GigabitEthernet0/1
AWL - L01	Wireless0	Wireless2.4g
AWL - L02	Wireless0	Wireless2.4g
AWL - L03	Wireless0	Wireless2.4g
AWL - L04	Wireless0	Wireless2.4g
AWL - L05	Wireless0	Wireless2.4g
	IOT – Access Point	
Device	Source Port	Access Point Port
IOT - Switch	Port0	GigabitEthernet0/1
IOT - L1	Wireless0	Wireless2.4g
IOT - S1	Wireless0	Wireless2.4g
IOT - M1	Wireless0	Wireless2.4g

iv. Logical Topology

LAB NO.	SUBNET DESCRIPTION	REQUIRED HOSTS	SUBNET MASK	USABLE HOST	NETWORK ADDRESS	1ST USABLE IP ADDRESS	LAST USABLE IP ADDRESS	BROADCAST ADDRESS
1	HUMAC	14	255.255.255.240 (/28)	14	192.168.10.96	192.168.10.97	192.168.10.110	192.168.10.111
2	SET - CL	7	255.255.255.224 (/27)	30	192.168.10.64	192.168.10.65	192.168.10.94	192.168.10.95
3	SET - AWL	10	255.255.255.224 (/27)	30	192.168.10.0	192.168.10.1	192.168.10.30	192.168.10.31
4	SET - IOT	13	255.255.255.224 (/27)	30	192.168.10.32	192.168.10.33	192.168.10.62	192.168.10.63
5	WAN	2	255.255.255.252 (/30)	2	192.168.10.112	192.168.10.113	192.168.10.114	192.168.10.115

Figure 3.1: Subnetting Table

The given network, 192.168.10.0/24 has been segmented into five subnets, with different subnet masks. In ***Figure 3.1***, each lab (i.e. CL, AWL, IOT, and HUMAC) has one subnet each, and the WAN link which uses routers to allow communication between the University building and SET building has one subnet. Each of them has their own network address and dedicated range of IP addresses that they are permitted to use.

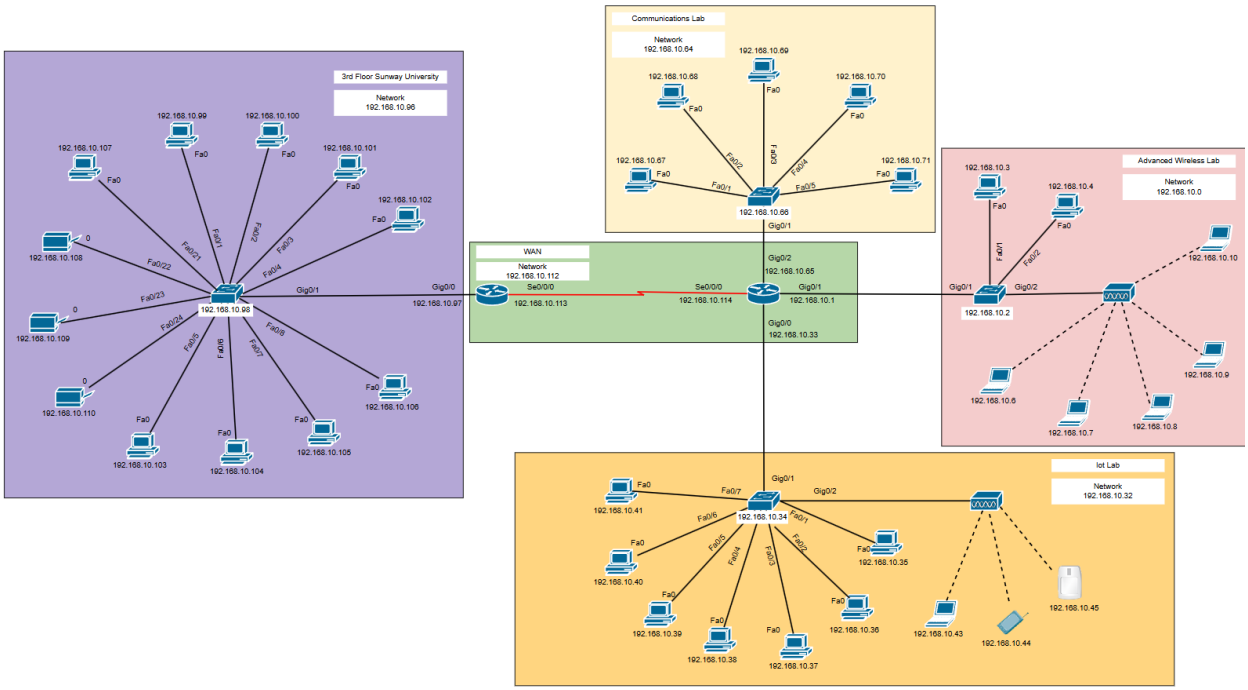


Figure 3.2: Logical Topology

Figure 3.2 shows the logical topology of the network. Each LAN and WAN has its own network address which has been assigned according to ***Figure 3.1***. Each device's IP address has been included as well.

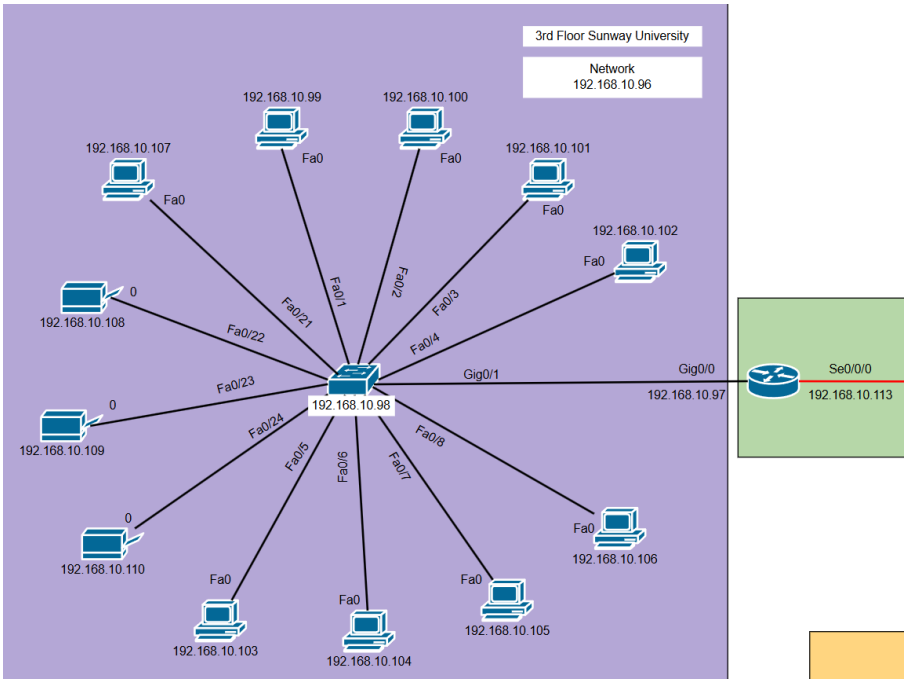


Figure 3.3: Logical Topology for HUMAC Lab

Figure 3.3 illustrates the logical topology of the HUMAC Lab that is located at the 3rd floor of Sunway University. The network address for this LAN is 192.168.10.96, with subnet mask 255.255.255.240 (/28 prefix) which has 14 usable host IP addresses available. The range of usable IP addresses for this LAN is from 192.168.10.97 to 192.168.10.110. Every end device in this LAN is connected to the Fast Ethernet Ports of the HUMAC Switch. Meanwhile, the Gigabit Ethernet port of the switch is used to connect to the HUMAC lab router.

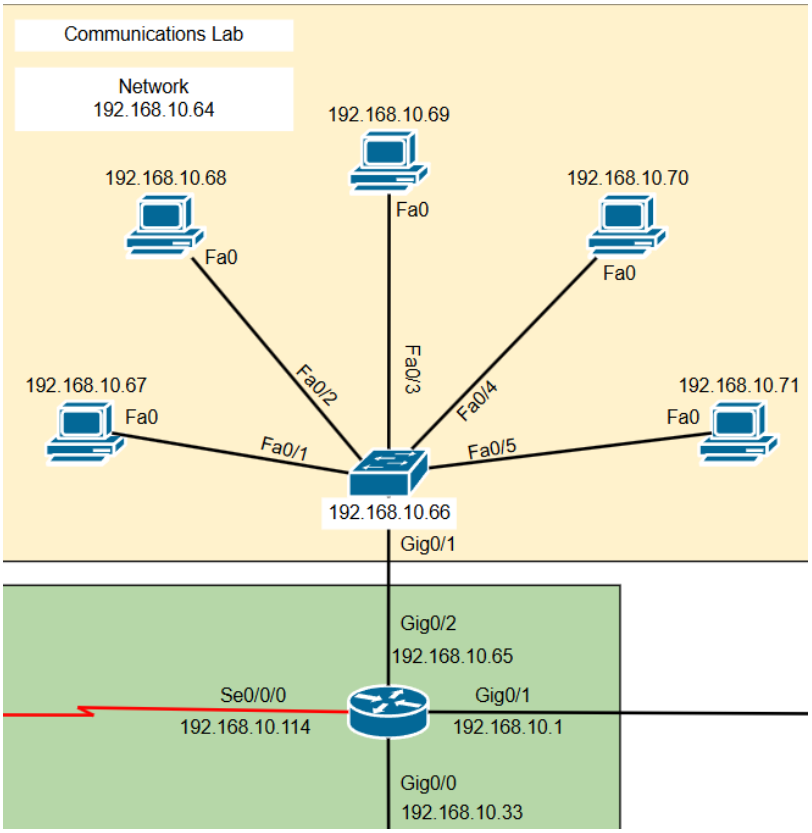


Figure 3.4: Logical Topology for CL Lab

Figure 3.4 illustrates the logical topology of the CL Lab that is located at the SET building. The

network address for this LAN is 192.168.10.64, with subnet mask 255.255.255.224 (/27 prefix) which has 30 usable host IP addresses available. The range of usable IP addresses for this LAN is from 192.168.10.65 to 192.168.10.94. Every end device in this LAN is connected to the Fast Ethernet Ports of the CL Switch. Meanwhile, the Gigabit Ethernet port of the switch is used to connect to the SET lab router.

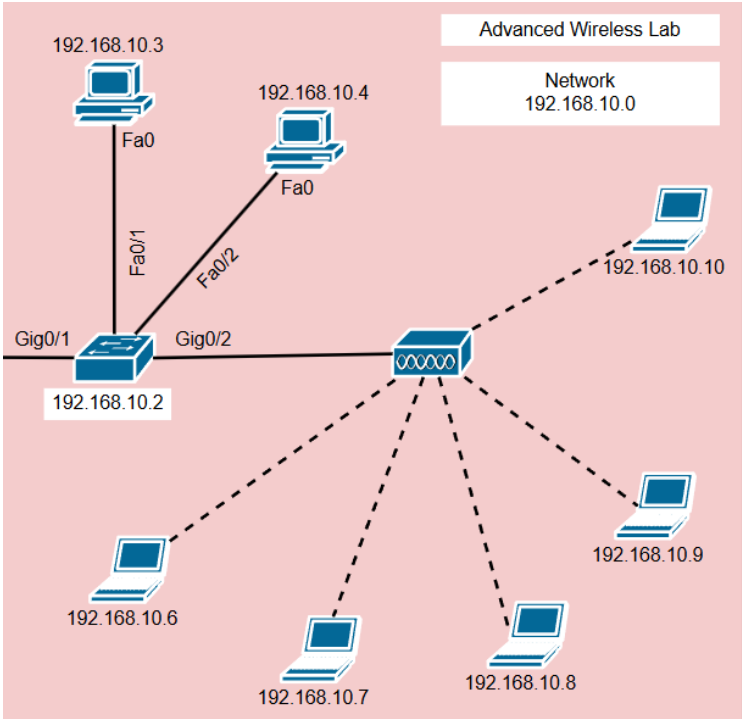


Figure 3.5: Logical Topology for AWL Lab

Figure 3.5 illustrates the logical topology of the AWL Lab that is located at the SET building. The network address for this LAN is 192.168.10.0, with subnet mask 255.255.255.224 (/27 prefix) which has 30 usable host IP addresses available. The range of usable IP addresses for this LAN is from 192.168.10.1 to 192.168.10.30. Every computer in this LAN is connected to the Fast Ethernet Ports of the AWL Switch, and wireless devices like the laptops are connected to the AWL Access Point. Meanwhile, the Gigabit Ethernet port of the switch is used to connect to the SET lab router and the AWL Access Point.

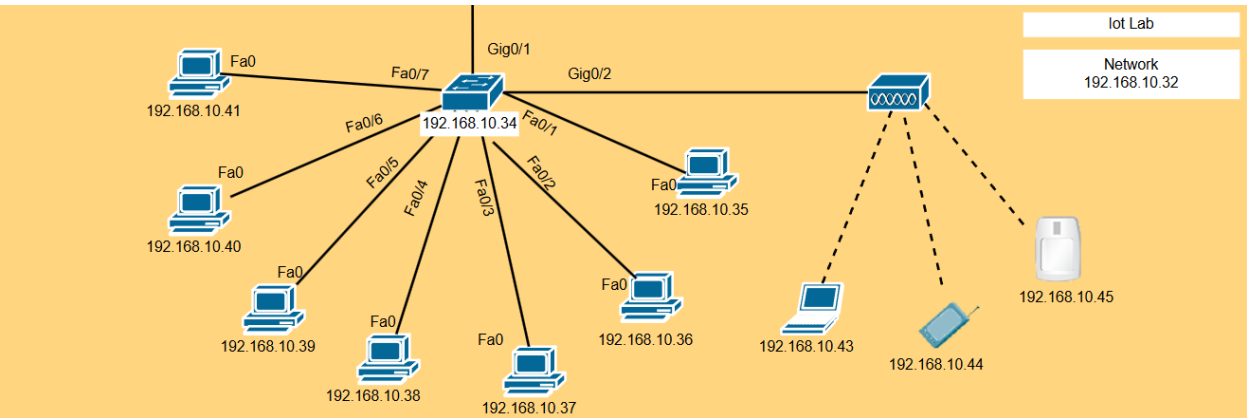


Figure 3.6: Logical Topology for IoT Lab

Figure 3.6 illustrates the logical topology of the IoT Lab that is located at the SET building. The network address for this LAN is 192.168.10.32, with subnet mask 255.255.255.224 (/27 prefix) which has 30 usable host IP addresses available. The range of usable IP addresses for this LAN is from 192.168.10.33 to 192.168.10.62. Every computer in this LAN is connected to the Fast

Ethernet Ports of the IoT Switch, and wireless devices like the laptop, smartphone, and motion detector are connected to the IoT Access Point. Meanwhile, the Gigabit Ethernet port of the switch is used to connect to the SET lab router and the IoT Access Point.

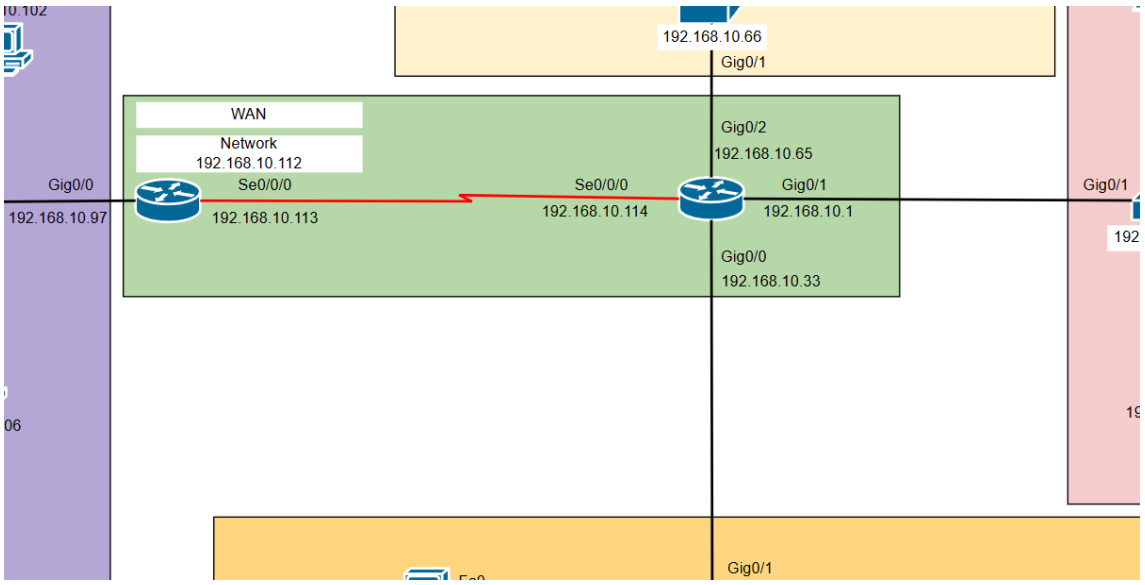


Figure 3.7: Logical Topology for the WAN Link

Figure 3.7 illustrates the logical topology of the WAN Link of the HUMAC router and SET Router that are located at the University building and SET building respectfully. The network address for this LAN is 192.168.10.112, with subnet mask 255.255.255.252 (/30 prefix) which has 2 usable host IP addresses available. The range of usable IP addresses for this LAN is from 192.168.10.113 to 192.168.10.114. The routers are connected through their serial ports, and their Gigabit Ethernet ports are used to connect to each lab's (i.e. HUMAC lab, CL lab, AWL lab, and IoT Lab) switch.

PART B: IMPLEMENTATION

i. Overview of Design

This network interconnects 4 labs within Sunway University that is located in Subang Jaya, Selangor, Malaysia. The primary objective of this network is to facilitate smooth communication and effective resource sharing across the HUMAC lab, Communications Lab (CL), the Advance Wireless Lab (AWL), and the IoT Lab (IOT). Specific security and management requirements are incorporated into this design, such as the requirement to accommodate varying host counts per lab, network segmentation for better control, and the use of the available network address space. The implementation of this network aims to provide and reliable communication infrastructure that is able to support the academic and research activities within these labs.

Sequential network addresses

5 subnets

SUBNET DESCRIPTION	REQUIRED HOSTS	SUBNET MASK	USABLE HOST	NETWORK ADDRESS	ST USABLE IP ADDRESS	LAST USABLE IP ADDRESS	BROADCAST ADDRESS
SET - AWL	10	255.255.255.224 (/27)	30	192.168.10.0	192.168.10.1	192.168.10.30	192.168.10.31
SET - IOT	13	255.255.255.224 (/27)	30	192.168.10.32	192.168.10.33	192.168.10.62	192.168.10.63
SET - CL	7	255.255.255.224 (/27)	30	192.168.10.64	192.168.10.65	192.168.10.94	192.168.10.95
HUMAC	14	255.255.255.240 (/28)	14	192.168.10.96	192.168.10.97	192.168.10.110	192.168.10.111
WAN	2	255.255.255.252 (/30)	2	192.168.10.112	192.168.10.113	192.168.10.114	192.168.10.115

Network Architecture:

This network design utilizes a **Point-to-Point Wide-Area Network (WAN) topology**. This network architecture comprises of 2 nodes (main routers) located at 2 sub-locations within Sunway University which are 3rd Floor, Sunway University, and the SET Building. The WAN link between these 2 routers enables connectivity between the host devices from different buildings within the Sunway campus. The **Ethernet WAN connection** directly connects these 2 nodes and extends the LAN access technology into the WAN such that the LAN of each lab will be extended to a WAN via routers and switches.

The network interconnects a total of 4 labs within Sunway University that utilize **star topology** within their Local Area Network (LAN). At the 3rd Floor of Sunway University lies the HUMAC lab which has 12 hosts in total. The SET building, on the other hand, has a total of 3 switches located respectively in the remaining 3 labs, i.e. the CL, AWL and IOT labs. Each lab is equipped with its own dedicated switch in order to improve the overall network performance by reducing unnecessary traffic on other parts of the network as well as allowing seamless scalability and flexibility, especially when there is a new host added into a certain lab. A wireless router is also incorporated within the AWL to support wireless connectivity for end devices such as laptops. This design ensures for efficient traffic flow and simplifies network management, not to mention, simplify troubleshooting by fault isolation within the respective lab switch.

The design of the network was influenced by certain key requirements which includes:

- <u>Host Capacity</u>
  - Each lab had been assigned a specific number of hosts (required hosts). Subnets of the network are selected based on the quantity of hosts it can support. For instance, the HUMAC lab on the 3rd Floor, Sunway University was assigned the /28 subnet which could accommodate up to 14 usable host per subnet.

- <u>Segmentation of Network</u>
  - The network is segmented into 5 subnets, one for each of the 4 labs and another for

the WAN link.

- Subnets are allocated based on the number of host addresses required for each network and the number of individual subnets needed. For instance, in the WAN link, only 2 host addresses are needed for each router. Hence, subnet /30, with 2 usable host addresses, is chosen as it only requires the smallest possible subnet for a point-to-point connection. This allows for improved management and security while maximizing the allocated subnets and minimizing unused hosts.

- In the labs, subnet masks with larger number of usable hosts are allocated to take into account of scalability, in the occurrence of adding additional host devices within the LAN. For example, in the AWL lab located at the SET building, subnet /27 that can accommodate up to 30 usable host is chosen although there are only 10 required hosts. This allow for easy and efficient expansion of the network, without affecting the other labs such that configurations and troubleshooting can be executed within each lab's network independently.

- <u>Network Address</u>
  - The network design utilizes the provided network address space of 192.168.10.0/24. This provides a range of IP addresses from 192.168.10.0 to 192.168.10.255.

  - The network addresses are chosen to be the first available address in each subsequent subnet range sequentially to prevent subnets of the 4 different labs and WAN link from overlapping.

Equipment Selection:
The equipment models within the network were chosen based on their availability in the market, cost-effectiveness and required functionality.

Security Considerations:
To enhance network security, password encryption is implemented on the main routers and switches. This measure protects sensitive configuration information and prevents unauthorized access to the network infrastructure. The password used to access the user EXEC mode for the routers and switches is **'user'**. Meanwhile, the password used to access the privileged EXEC mode is '**priv**'.

Subnetting Scheme:
The assigned network, which is 192.168.10.0/24, is subnetted into five parts. The University building has one subnet for the HUMAC Lab LAN, the network address allocated for it is 192.168.10.96. It takes the subnet mask 255.255.255.240 (/28) as it provides 14 usable hosts, which satisfy the required number of hosts for the HUMAC Lab.

The SET building has three subnets, one for each lab which are the Communication Lab (CL), Advanced Wireless Lab (AWL), and IoT Lab (IOT). Network address for CL Lab is 192.168.10.64, AWL is 192.168.10.0, and IOT is 192.168.10.32. All have the subnet mask 255.255.255.224 (/27) which provides 30 usable hosts. The WAN link has one subnet allocated to it, with the network address of 192.168.10.112 and subnet mask of 255.255.255.252 (/30).

IP Addressing Scheme:
All devices on the network have their IP addresses assigned statically and according to the IPv4 addressing scheme. The addressing table has been coloured coded for easy reference, beige for the CL LAN, pink for the AWL LAN, orange for the IOT LAN, purple for the HUMAC LAN, and green for the WAN link.

HUMAC
SET - CL
SET - AWL
SET - IOT
WAN

Figure 4.1: Colour Codes for each LANs and WAN

The following table illustrates the IP addresses assigned to each device on the network, also known as the addressing table.

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
HUMAC Router	G0/0 (to HUMAC switch)	192.168.10.97	255.255.255.240	N.A.
	Serial0/0/0 (to SET router)	192.168.10.113	255.255.255.252	N.A.
SET router	G0/2 (to CL switch)	192.168.10.65	255.255.255.224	N.A.
	G0/1 (to AWL switch)	192.168.10.1	255.255.255.224	N.A.
	G0/0 (to IOT switch)	192.168.10.33	255.255.255.224	N.A.
	Serial0/0/0 (to HUMAC router)	192.168.10.114	255.255.255.252	N.A.
HUMAC - Switch	VLAN1	192.168.10.98	255.255.255.240	192.168.10.97
CL - Switch	VLAN1	192.168.10.66	255.255.255.224	192.168.10.65
AWL - Switch	VLAN1	192.168.10.2	255.255.255.224	192.168.10.1
IOT - Switch	VLAN1	192.168.10.34	255.255.255.224	192.168.10.33
3A - 01	NIC (FastEthernet0/1)	192.168.10.99	255.255.255.240	192.168.10.97
3A - 02	NIC (FastEthernet0/2)	192.168.10.100	255.255.255.240	192.168.10.97
3A - 03	NIC (FastEthernet0/3)	192.168.10.101	255.255.255.240	192.168.10.97
3A - 04	NIC (FastEthernet0/4)	192.168.10.102	255.255.255.240	192.168.10.97
3B - 01	NIC (FastEthernet0/5)	192.168.10.103	255.255.255.240	192.168.10.97
3B - 02	NIC (FastEthernet0/6)	192.168.10.104	255.255.255.240	192.168.10.97
3B - 03	NIC (FastEthernet0/7)	192.168.10.105	255.255.255.240	192.168.10.97
3B - 04	NIC (FastEthernet0/8)	192.168.10.106	255.255.255.240	192.168.10.97
Printer Room PC	NIC (FastEthernet0/21)	192.168.10.107	255.255.255.240	192.168.10.97
Printer 1	NIC (FastEthernet0/22)	192.168.10.108	255.255.255.240	192.168.10.97

Printer 2	NIC (FastEthernet0/23)	192.168.10.109	255.255.255.240	192.168.10.97
Printer 3	NIC (FastEthernet0/24)	192.168.10.110	255.255.255.240	192.168.10.97
CL - 01	NIC (FastEthernet0/1)	192.168.10.67	255.255.255.224	192.168.10.65
CL - 02	NIC (FastEthernet0/2)	192.168.10.68	255.255.255.224	192.168.10.65
CL - 03	NIC (FastEthernet0/3)	192.168.10.69	255.255.255.224	192.168.10.65
CL - 04	NIC (FastEthernet0/4)	192.168.10.70	255.255.255.224	192.168.10.65
CL - 05	NIC (FastEthernet0/5)	192.168.10.71	255.255.255.224	192.168.10.65
AWL - L01	NIC (Wireless2.4g)	192.168.10.6	255.255.255.224	192.168.10.1
AWL - L02	NIC (Wireless2.4g)	192.168.10.7	255.255.255.224	192.168.10.1
AWL - L03	NIC (Wireless2.4g)	192.168.10.8	255.255.255.224	192.168.10.1
AWL - L04	NIC (Wireless2.4g)	192.168.10.9	255.255.255.224	192.168.10.1
AWL - L05	NIC (Wireless2.4g)	192.168.10.10	255.255.255.224	192.168.10.1
AWL - PC1	NIC (FastEthernet0/1)	192.168.10.3	255.255.255.224	192.168.10.1
AWL - PC2	NIC (FastEthernet0/2)	192.168.10.4	255.255.255.224	192.168.10.1
IOT - PC1	NIC (FastEthernet0/7)	192.168.10.35	255.255.255.224	192.168.10.33
IOT - PC2	NIC (FastEthernet0/6)	192.168.10.36	255.255.255.224	192.168.10.33
IOT - PC3	NIC (FastEthernet0/5)	192.168.10.37	255.255.255.224	192.168.10.33
IOT - PC4	NIC (FastEthernet0/4)	192.168.10.38	255.255.255.224	192.168.10.33
IOT - PC5	NIC (FastEthernet0/3)	192.168.10.39	255.255.255.224	192.168.10.33
IOT - PC6	NIC (FastEthernet0/2)	192.168.10.40	255.255.255.224	192.168.10.33
IOT - PC7	NIC (FastEthernet0/1)	192.168.10.41	255.255.255.224	192.168.10.33
IOT - L1	NIC (Wireless2.4g)	192.168.10.43	255.255.255.224	192.168.10.33
IOT - S1 (smartphone)	NIC	192.168.10.44	255.255.255.224	192.168.10.33
IOT - M1 (motion detector)	NIC	192.168.10.45	255.255.255.224	192.168.10.33

ii. Configuration

End Devices Configuration (Computers, Laptops, Printers)

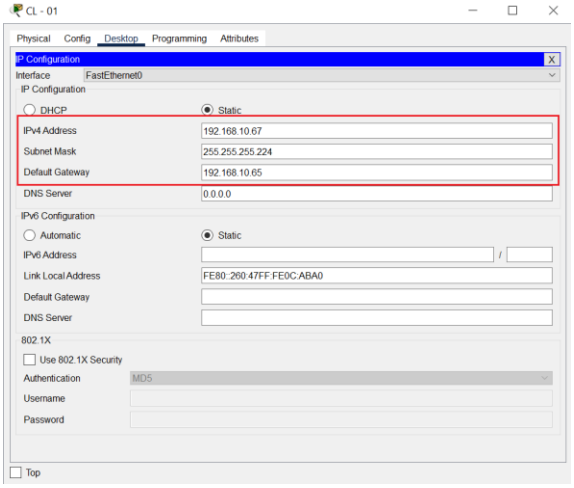


Figure 5.1: Computer Configuration

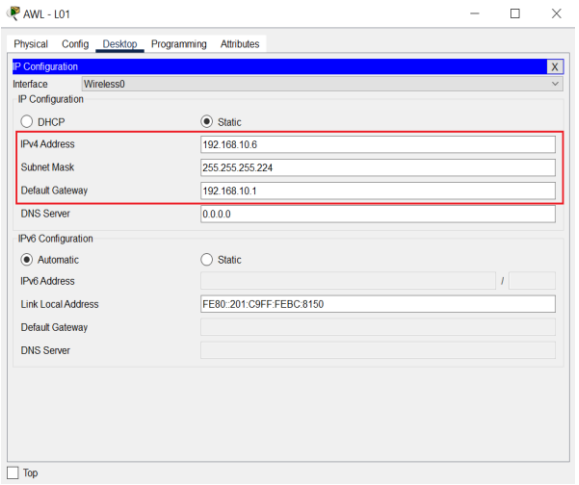


Figure 5.2: Laptop Configuration

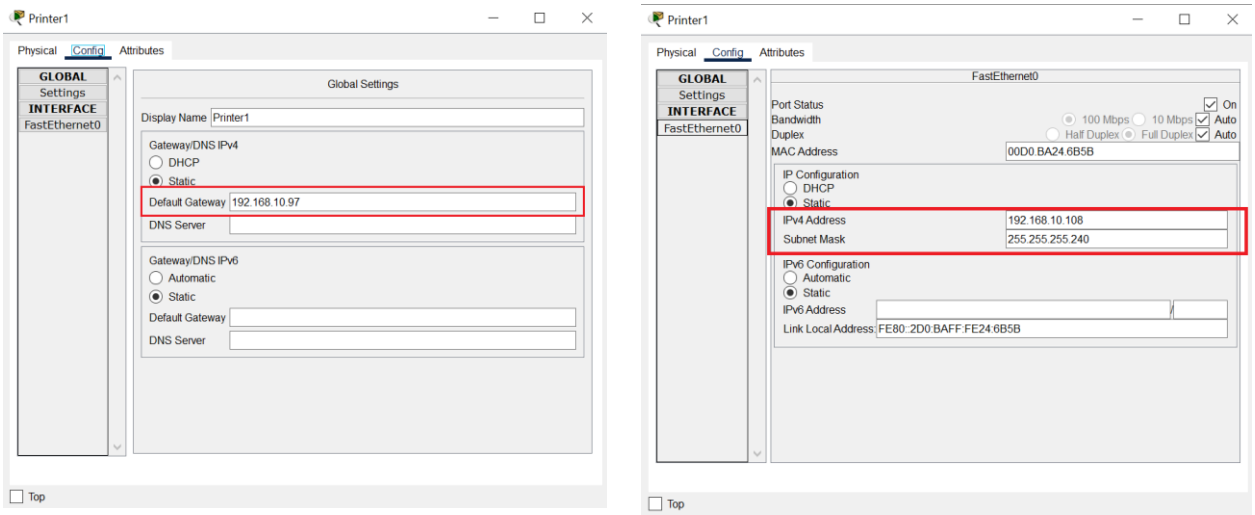


Figure 5.3: Printer Configuration

Each end device will have their IPv4 addresses, subnet mask, and default gateway configured statically based on the IP addressing table created. ***Figures 5.1, 5.2*** and ***5.3*** shows the configuration of computer CL-01 from the CL lab, laptop AWL-01 from the AWL lab, and printer Printer1 from the HUMAC lab as examples of how these end devices in the network are configured.

Wireless End Devices Configuration (IoT Devices)

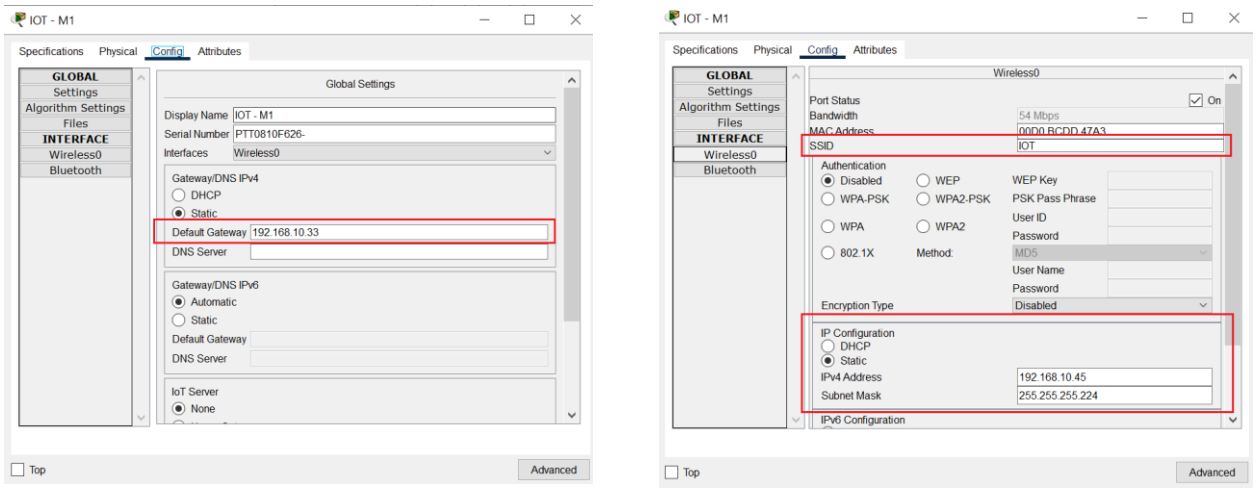


Figure 5.4: IoT Device Configuration

Figure 5.4 shows the configuration of motion detector IOT-M1 from the IOT lab as an example of how the IoT devices in this lab are configured.

Switch Configuration

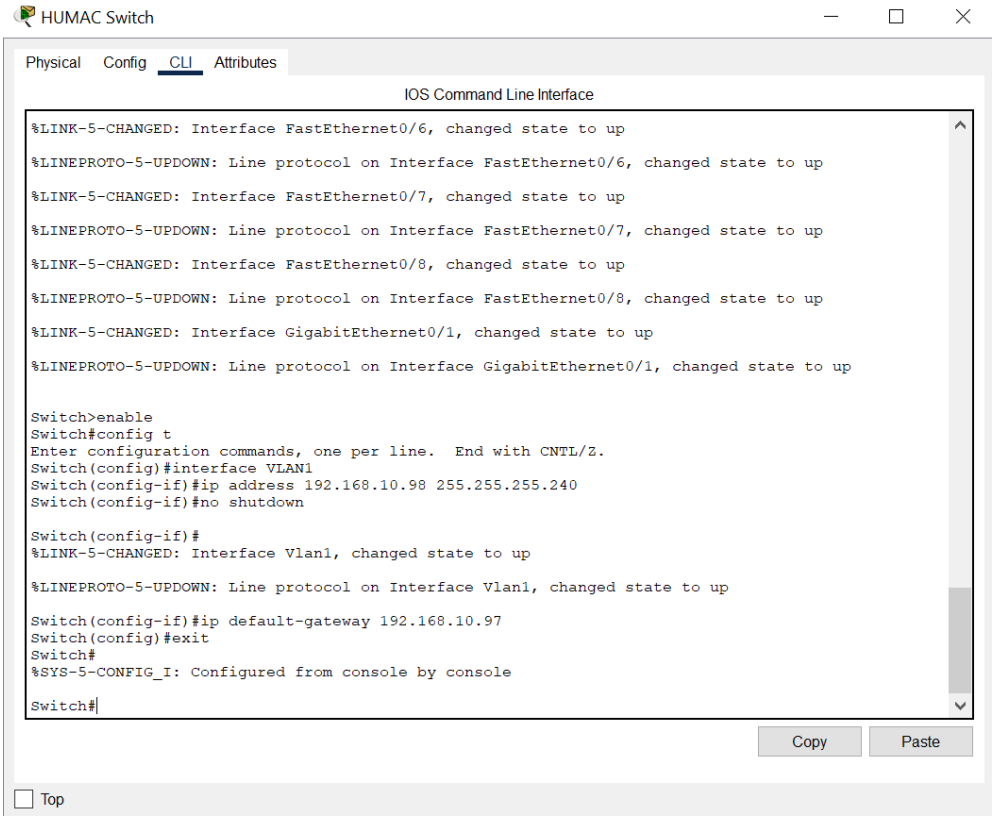


Figure 5.5: Switch Configuration

Figure 5.5 shows the configuration of the HUMAC Switch from the HUMAC lab as an example of how all switches in this network are configured (i.e. HUMAC Switch, CL Switch, AWL Switch, and IOT Switch). The 'enable' command allows access to the privileged EXEC mode, entering the 'config t' command in this mode will then change the mode to the global configuration mode.

The command ‘interface VLAN1’ is entered which will allow configurations to the VLAN1 interface. From there, the IP address and subnet mask can be configured according to the IP addressing table created. In this example, the IP address assigned for the HUMAC switch is 192.168.10.98, and 255.255.255.240 for the subnet mask. Then, the ‘no shutdown’ command is entered to activate the interface. After that, the default gateway is entered by using the ‘ip default-gateway' command. The IP address entered for this command is the router’s IP address. Referring to Figure 5.5, that would be 192.168.10.97, which is the IP address of the HUMAC Lab Router.

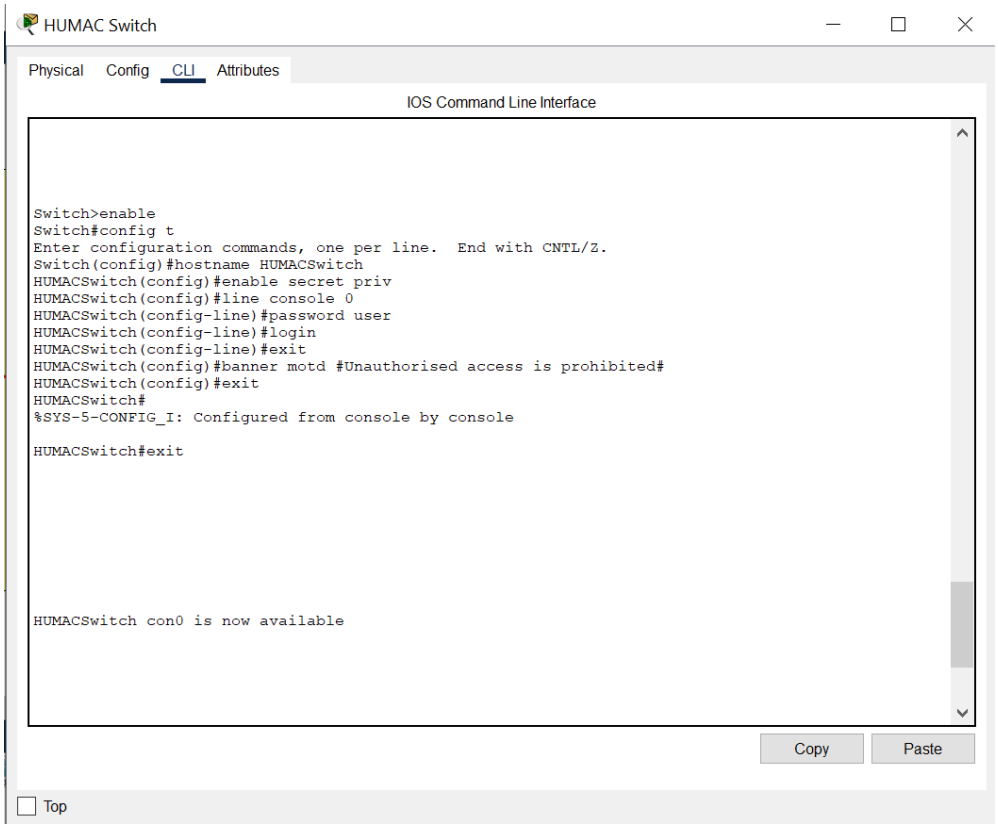


Figure 5.6: Security Configuration

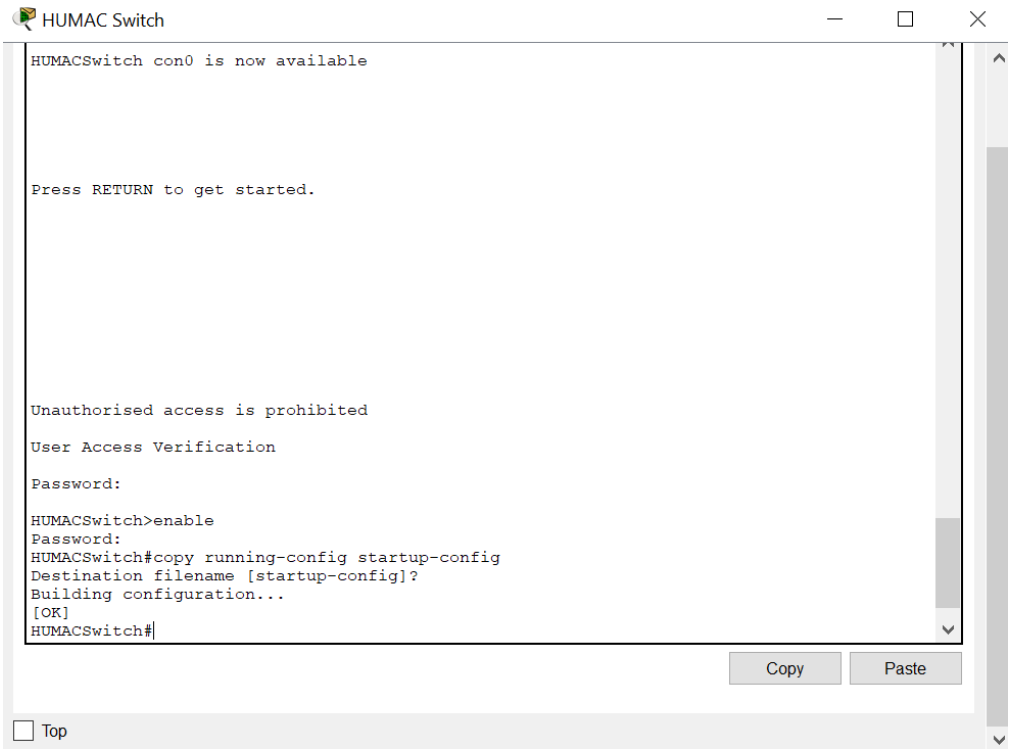


Figure 5.7: Demonstration of Accessing Various Modes

Global configuration mode is accessed first in order to change the hostname to 'HUMACSwitch' by using the 'hostname' command. To prevent unauthorised access to the switch, several passwords are used. The first one is to secure the user EXEC mode by requiring the user to login using a password. After entering the interface command mode for VLAN1 using the command 'interface VLAN1', the password is set to 'user' using the 'password' command.

The second one is to secure the privilege EXEC mode by requiring the user to enter a password, and this password is set to 'priv' by using the command 'enable secret'. Additionally, a banner message is implemented using the 'banner motd' command in the global configuration mode. The message set is 'Unauthorised access is prohibited' which appears upon login of the device. This serves as a legal warning to deter any unauthorised personnel from attempting to access the switch's terminal. Once the configurations have been set, using the 'copy running-config startup-config' saves the configurations to NVRAM.

Router Configuration

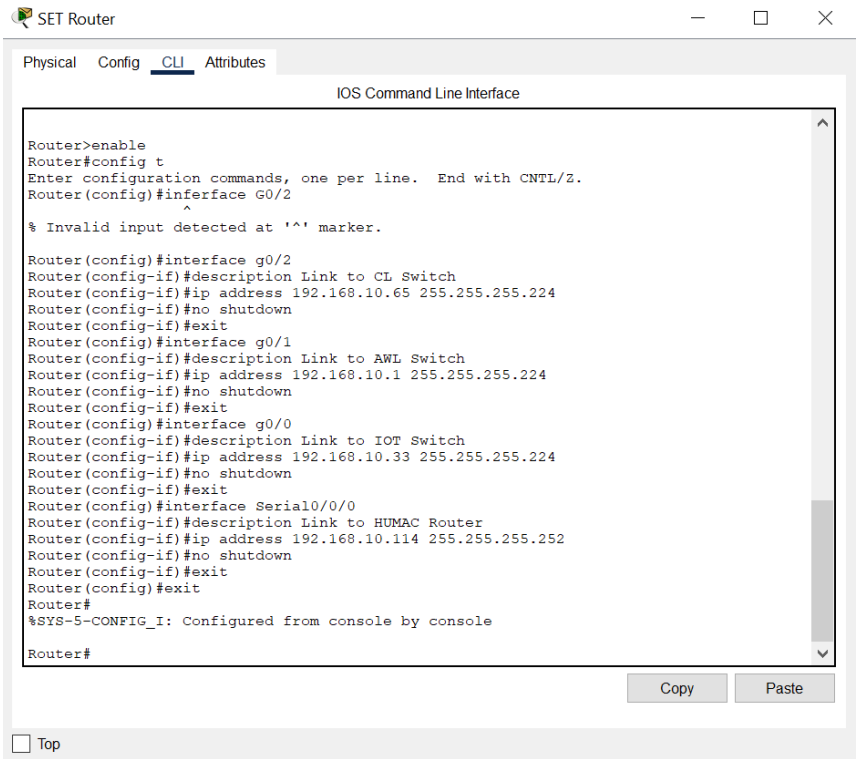


Figure 5.8: Router Configuration

Figure 5.8 shows the configuration of the SET Router from the SET building as an example of how all routers in this network are configured (i.e. SET Router and HUMAC Router). Once the global configuration mode is accessed, each connecting interface on the router will have its IP address and subnet mask configured (i.e. g0/2, g0/1, g0/0, and Serial0/0/0). The interface is activated by issuing the 'no shutdown' command. The 'description' command is used to add information about interfaces, which will help during troubleshooting when there is a lot of interfaces involved.

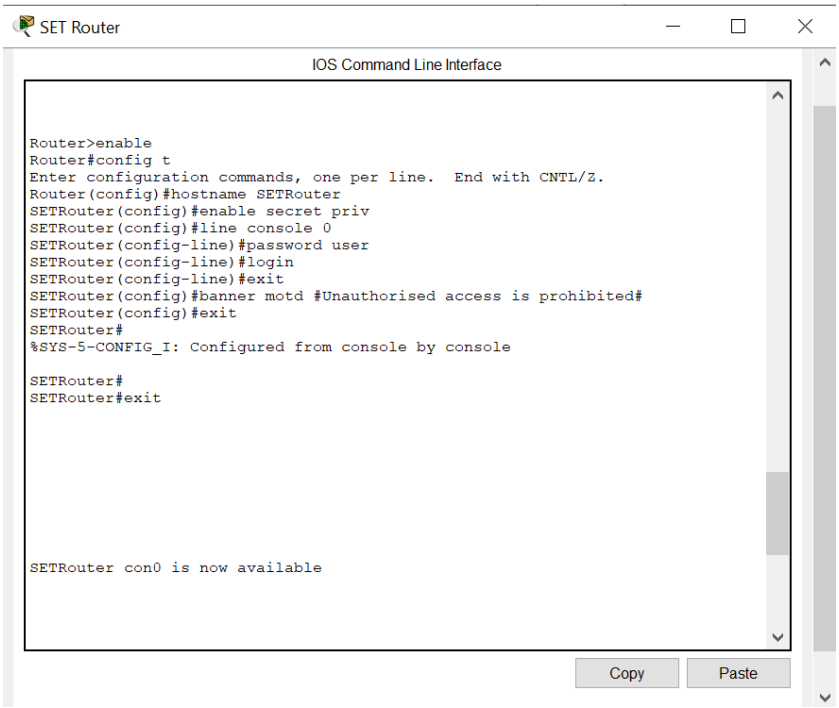


Figure 5.9: Security Configuration

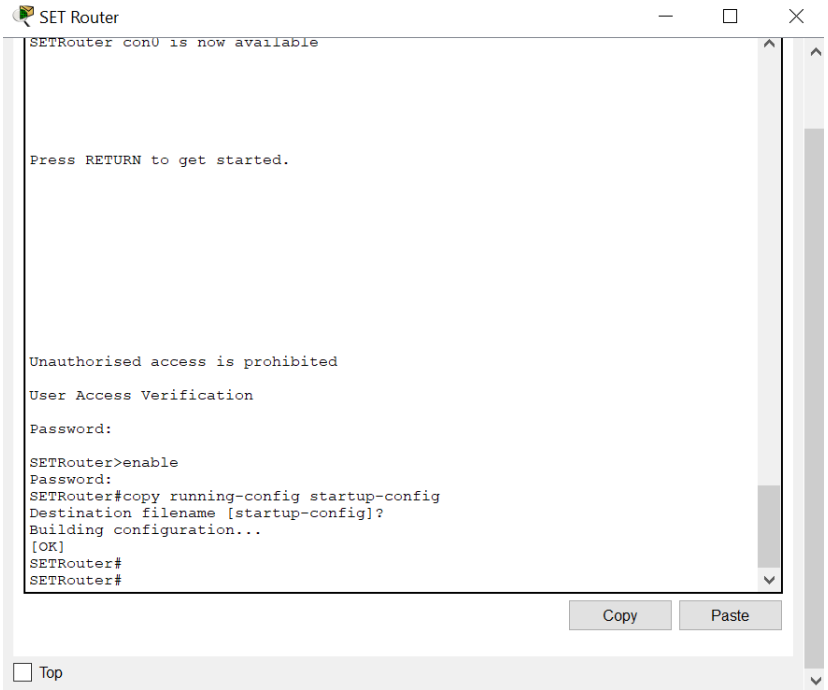


Figure 5.10: Demonstration of Accessing Different Nodes

In ***Figure 5.10***, the hostname is changed to SETRouter by using the 'hostname' command. A banner message appears when the interface is accessed for the first time, informing and deterring unauthorised personnel from tampering with the interface. The user EXEC mode is secured with a password 'user', privileged EXEC mode is secured with password 'priv'. Once the security configurations are set, the 'copy running-config startup-config' command saves the configurations.

Access Point Configuration

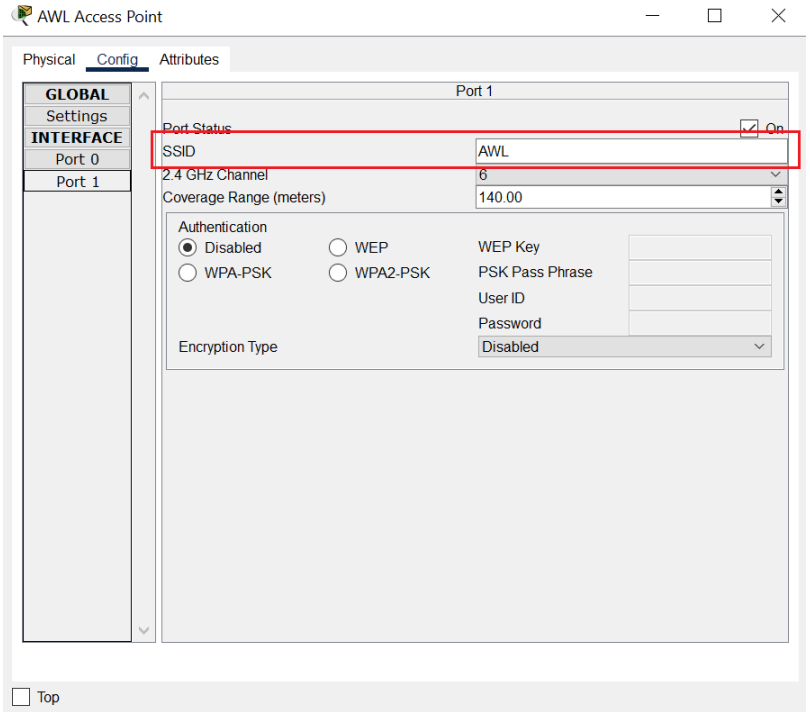


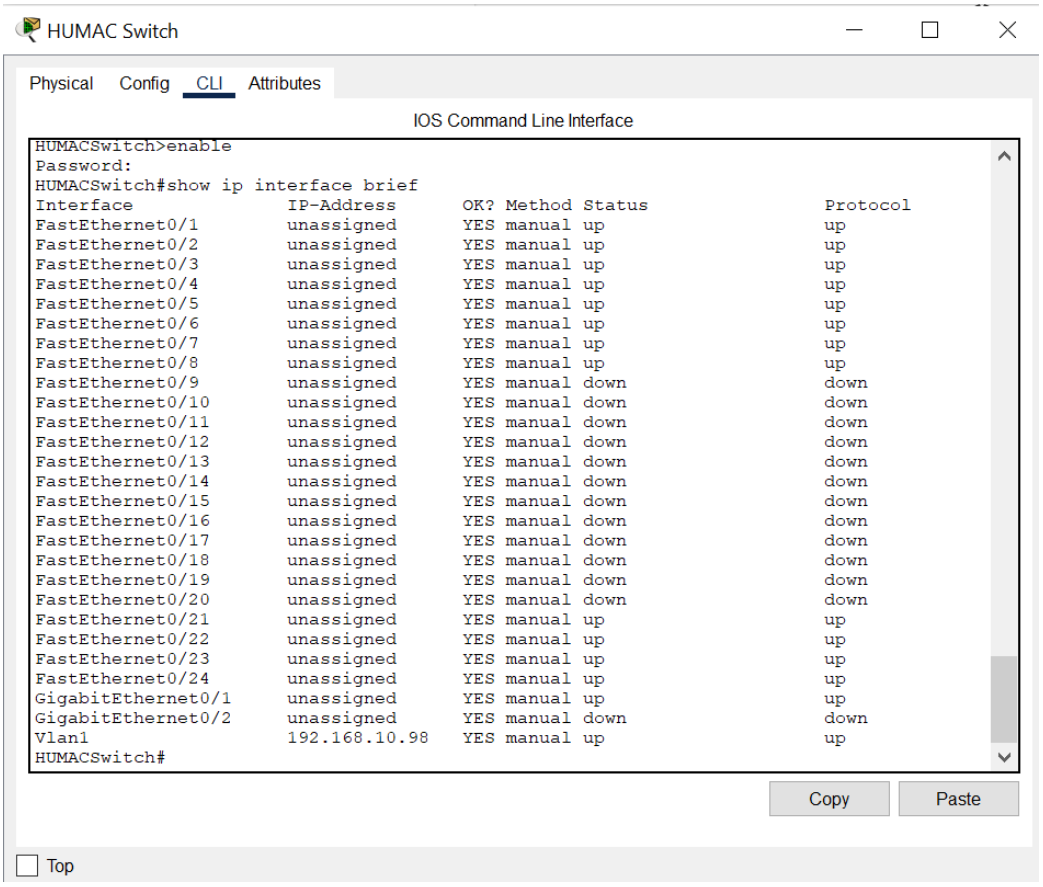
Figure 5.11: AWL Access Point Configuration

Figure 5.11 shows the configuration of the AWL Access Point from the AWL lab as an example of how all access points in this network are configured (i.e. AWL Access Point and IOT Access Point). The SSID (service set identifier) is a unique identifier that allows wireless devices to identify which network to connect to wirelessly. In the example, the SSID for the AWL Access Point has been configured to 'AWL'. Wireless end devices that wish to connect with the Access Point will have to have the SSID part configured to the SSID of the Access Point.

iii. Configuration Validation (Verifying Configuration of Devices)

The validation of the configuration of devices can be viewed with the command 'show ip interface brief'. The command will list down interfaces of the device along with their assigned IP address, and whether the interface is activated or not.

<u>Switch Configuration</u>




```
HUMACSwitch>enable
Password:
HUMACSwitch#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual up                    up
FastEthernet0/4        unassigned      YES manual up                    up
FastEthernet0/5        unassigned      YES manual up                    up
FastEthernet0/6        unassigned      YES manual up                    up
FastEthernet0/7        unassigned      YES manual up                    up
FastEthernet0/8        unassigned      YES manual up                    up
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual up                    up
FastEthernet0/22       unassigned      YES manual up                    up
FastEthernet0/23       unassigned      YES manual up                    up
FastEthernet0/24       unassigned      YES manual up                    up
GigabitEthernet0/1     unassigned      YES manual up                    up
GigabitEthernet0/2     unassigned      YES manual down                  down
Vlan1                  192.168.10.98   YES manual up                    up
HUMACSwitch#
```

Figure 6.1: IP Interface of HUMAC Switch

Figure 6.1 shows the HUMAC Switch's IP interface as an example of how the IP interface of switches (i.e. HUMAC Switch, AWL Switch, IOT Switch, and CL Switch) generally look in the network. Only the VLAN1 interface has been configured for switches, and in this example, the IP address for VLAN1 of the HUMAC Switch has been set to 192.168.10.98 according to the IP addressing table created.

Router Configuration

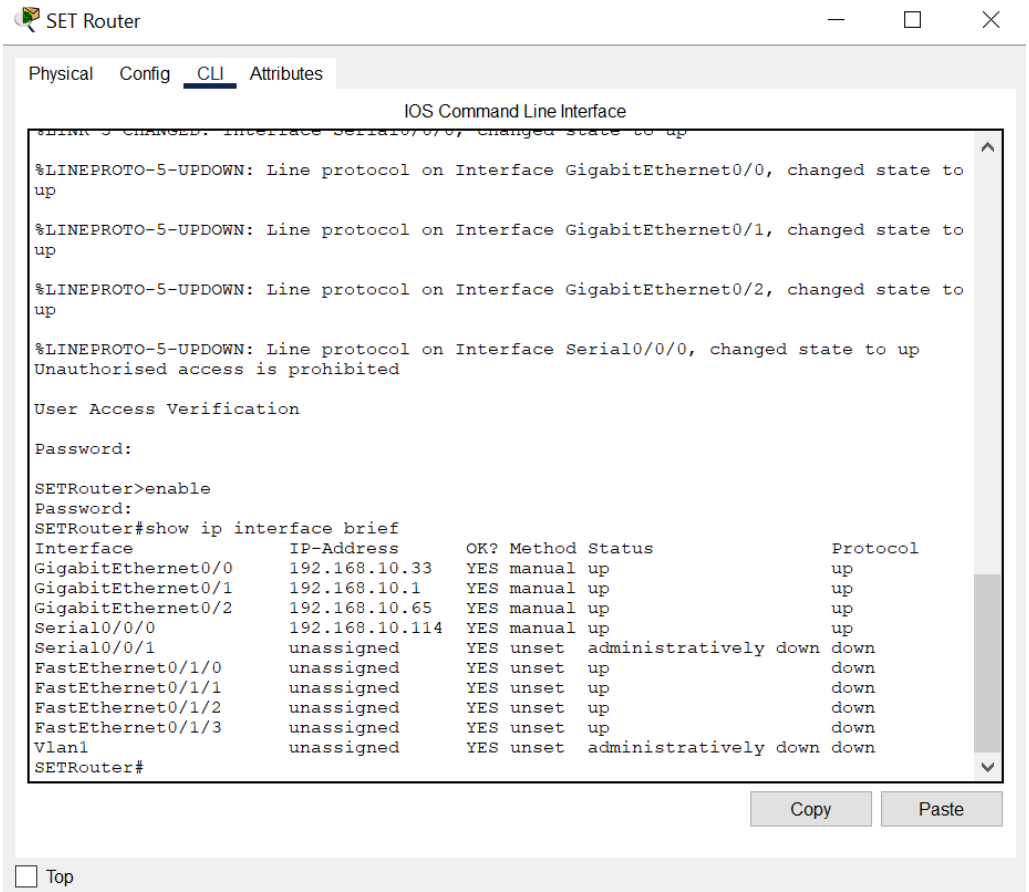


Figure 6.2: IP Interface of SET Router

Figure 6.2 shows the SET Router's IP interface as an example of how the IP interface of routers (i.e. SET Router and HUMAC Router) generally look in the network. G0/0 has the IP address of 192.168.10.33 which connects to the IOT Switch. G0/1 has the IP address of 192.168.10.1 which connects to the AWL Switch. G0/2 has the IP address of 192.168.10.65 which connects to the CL Switch. Serial 0/0/0 has the IP address of 192.168.10.114 which is part of the WAN link that connects the SET and HUMAC routers.

iv. Troubleshooting (Testing the Connectivity within the Network)

To test the connectivity of each device on the network, the command 'ping' followed by the destination IP address is used.

<u>Connectivity of Router to Switch</u>

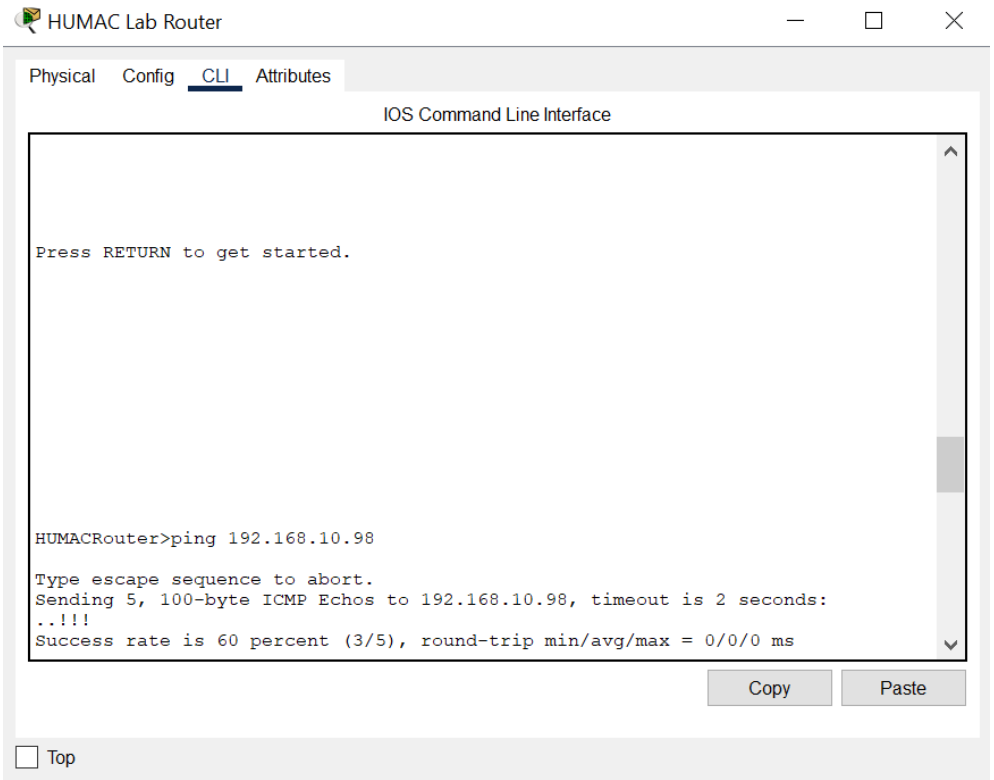


Figure 7.1: First Ping from HUMAC Router to HUMAC Switch

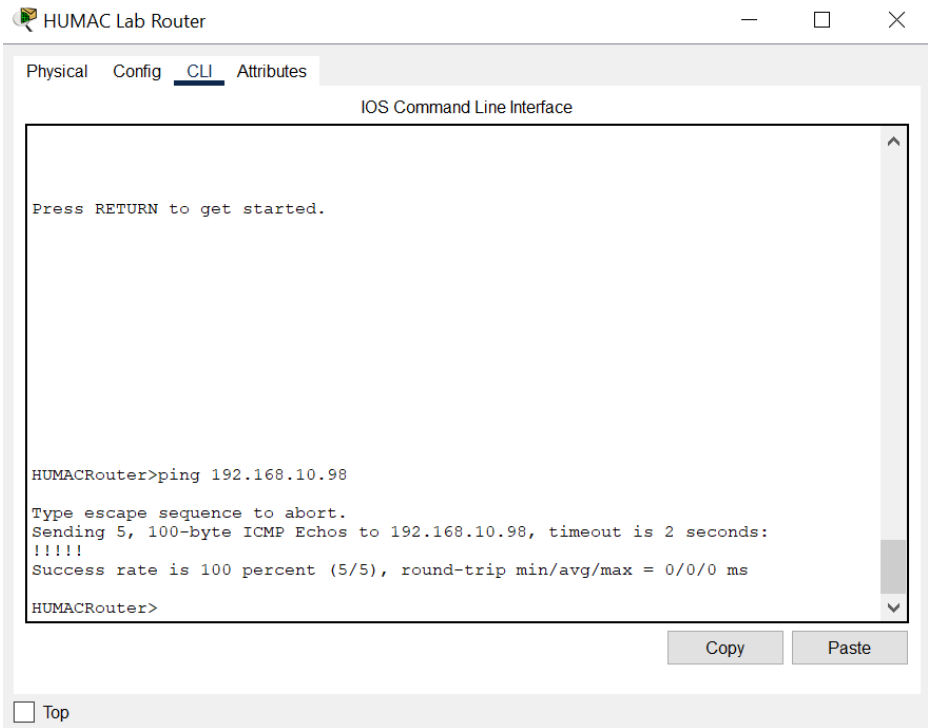


Figure 7.2: Second Ping from HUMAC Router CLI to HUMAC Switch

Figures 7.1 and ***7.2*** shows the testing of connectivity between the HUMAC Router and HUMAC Switch that are located in the University building. In the HUMAC Router's CLI, the IP address of

the HUMAC Switch (192.168.10.98) is pinged for two times. The first ping usually fails (has a success rate of 60 percent, meaning that two packets were dropped) because the router is not aware of the MAC address of the switch, so an ARP broadcast is sent. Once the MAC address of the switch is known and added to the ARP table, the success rate will be a 100 percent during the second ping, meaning that all packets are sent and received.

Connectivity of Switch to Switch

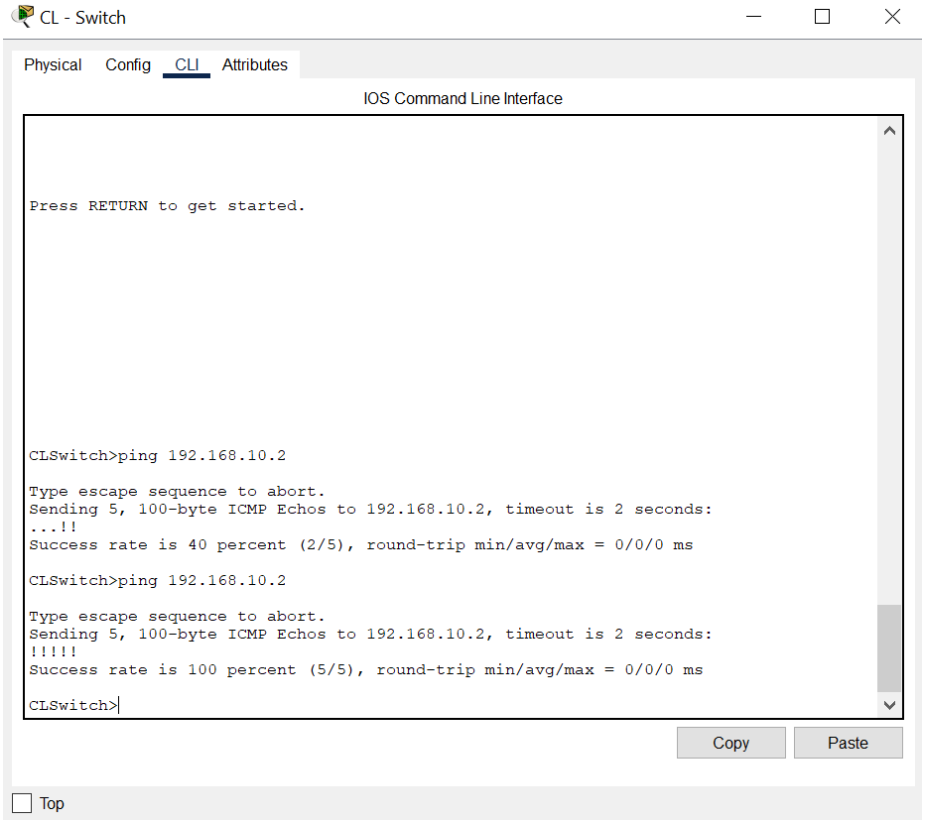


Figure 7.3: First Ping from CL Switch to AWL Switch

Figure 7.3 shows the testing of connectivity between CL Switch and AWL Switch, which are located in the CL lab and AWL lab respectfully. In the CLI of the CL Switch, the IP address of the AWL Switch (192.168.10.2) is pinged two times. During the first ping, three packets were dropped. After the first ping, packets are all send and received successfully.

Connectivity of End Devices within the same lab

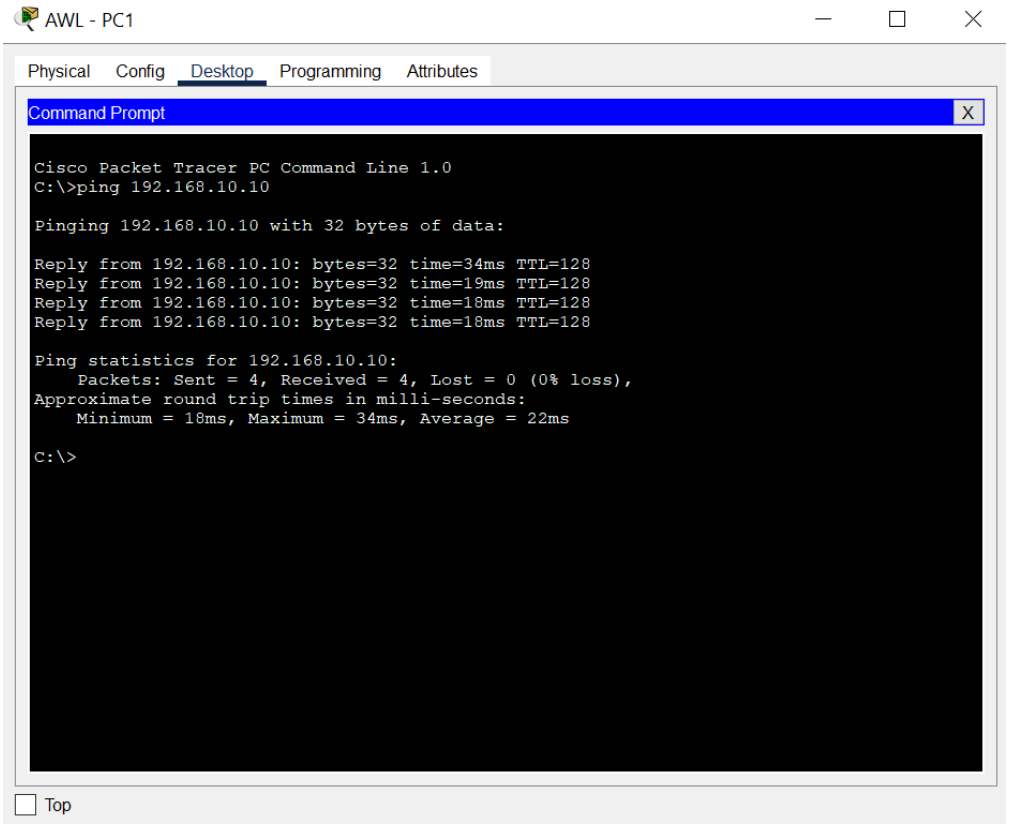


Figure 7.4: Ping between hosts (from AWL PC to AWL Laptop)

Figure 7.4 shows the testing of connectivity between computer AWL - PC1 and laptop AWL – L05 which are both located in the AWL lab. PC1 pings L05 with its IP address (192.168.10.10), and all packets were sent and received successfully.

Connectivity of End Devices in different buildings

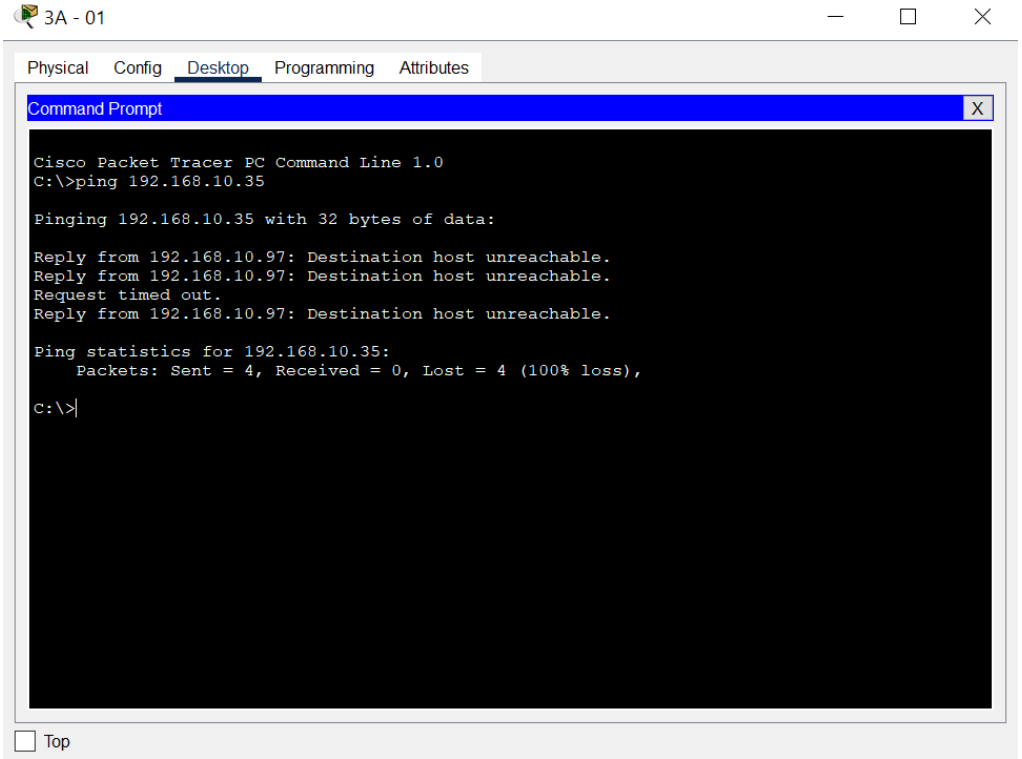


Figure 7.5: First ping between hosts (from HUMAC PC to IOT PC)

Figure 7.5 shows the testing of connectivity between computer 3A – 01 and computer IOT – PC01 in the HUMAC lab and IOT lab respectively, both labs are in different buildings. When 3A – 01 attempts to ping PC01 with its IP address (192.168.10.35), all packets are lost. The message 'Destination host unreachable' appears because the static routes have not been configured yet. This means that the routers are not aware of the path to deliver packets since the routing table does not have the information about the other routers. To solve this issue, the static routes for each router need to be configured manually.

The SET router would only need one static route configured, because there is only one lab in the University building, which is the HUMAC lab. Meanwhile, the HUMAC router would need three static routes configured, one route that leads to the AWL lab, one route that leads to the IOT lab, and one that leads to the CL lab.

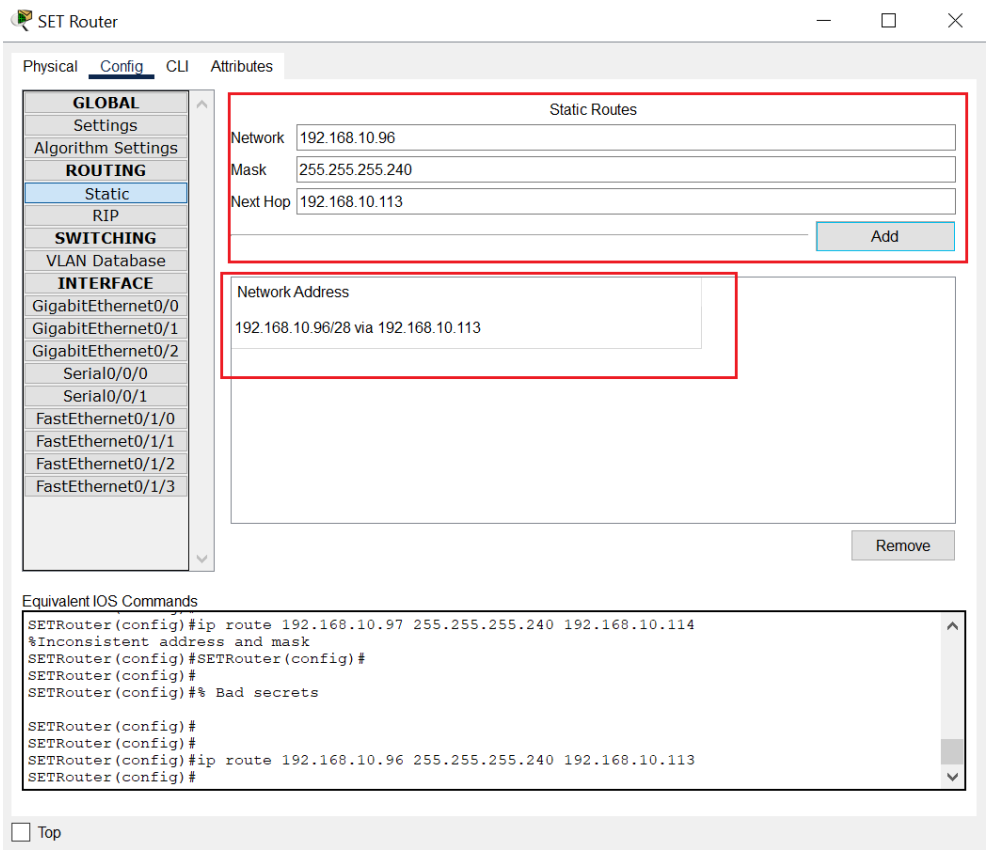


Figure 7.6: Configuring Static Route for SET Router

Figure 7.6 shows the configuration for the static route for the SET router. The IP address 192.168.10.96 is the network address of the HUMAC subnet, and the mask 255.255.255.240 also belongs to the HUMAC lab. The 'next hop' is the IP address of the Serial0/0/0 interface of the HUMAC router. Now, the SET router is aware of the route to direct messages from the SET building to the HUMAC lab.

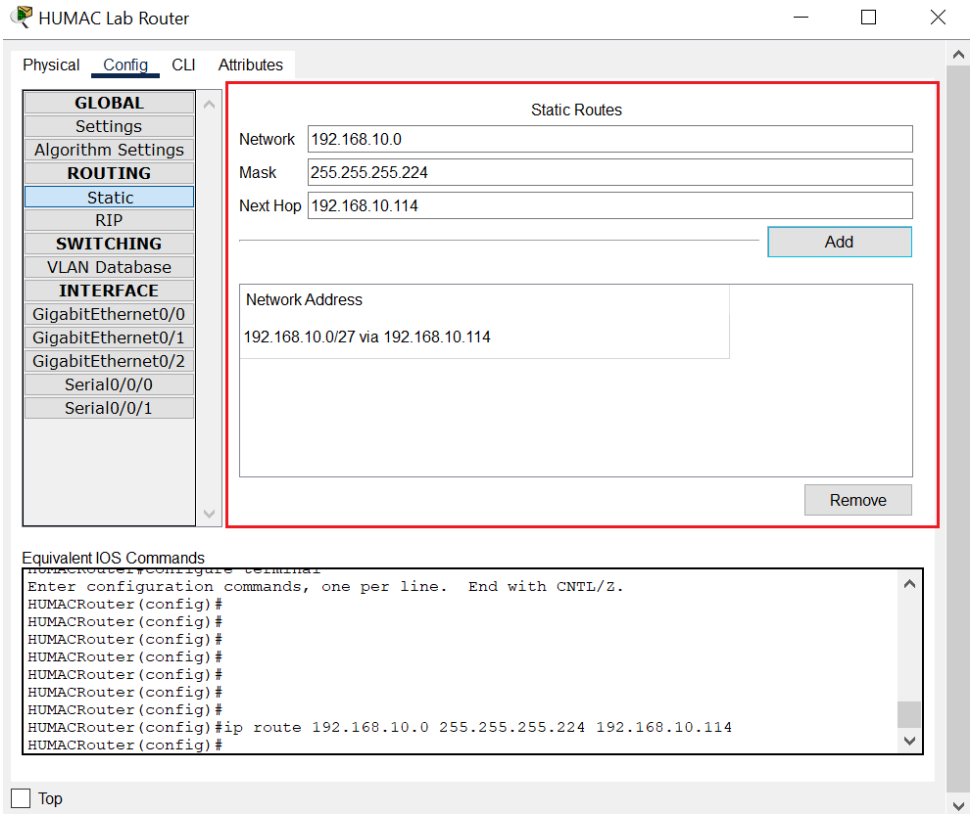


Figure 7.7: Configuring static route of HUMAC router (HUMAC Lab to AWL Lab)

Figure 7.7 shows the configuration for the first static route for the HUMAC router. This route is from the HUMAC lab to the AWL lab. The IP address 192.168.10.0 is the network address of the AWL subnet, and the mask 255.255.255.224 also belongs to the AWL subnet. The 'next hop' is the IP address of the Serial0/0/0 interface of the SET router. Now, the HUMAC router is aware of the route to direct messages from the HUMAC lab to the AWL lab.

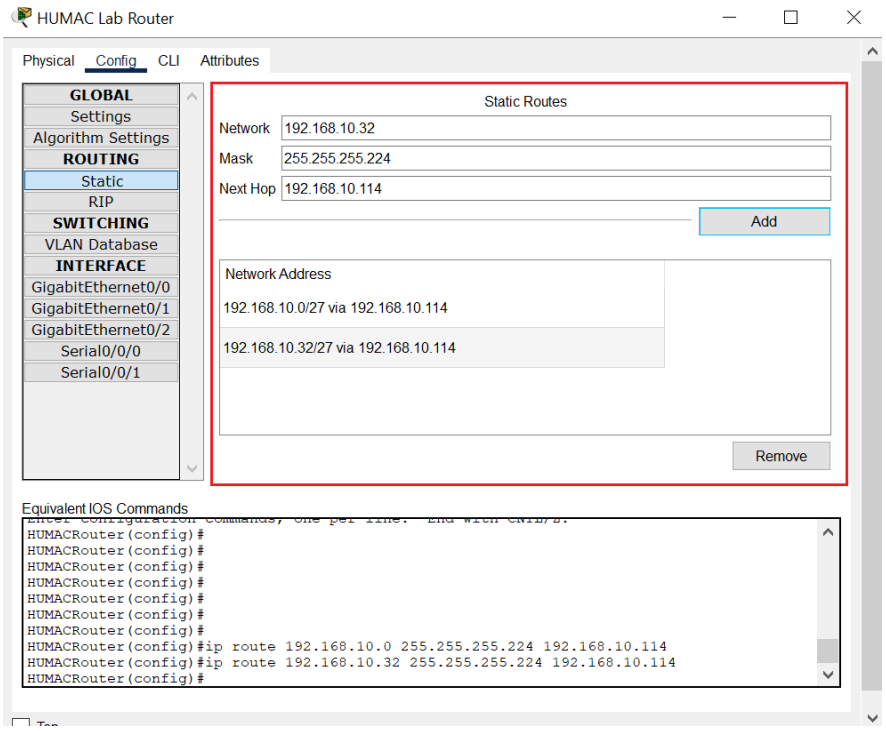


Figure 7.8: Configuring static route of HUMAC router (HUMAC Lab to IOT Lab)

Figure 7.8 shows the configuration for the second static route for the HUMAC router. This route is from the HUMAC lab to the IOT lab. The IP address 192.168.10.32 is the network address of

the IOT subnet, and the mask 255.255.255.224 also belongs to the IOT subnet. The 'next hop' is the IP address of the Serial0/0/0 interface of the SET router. Now, the HUMAC router is aware of the route to direct messages from the HUMAC lab to the IOT lab.

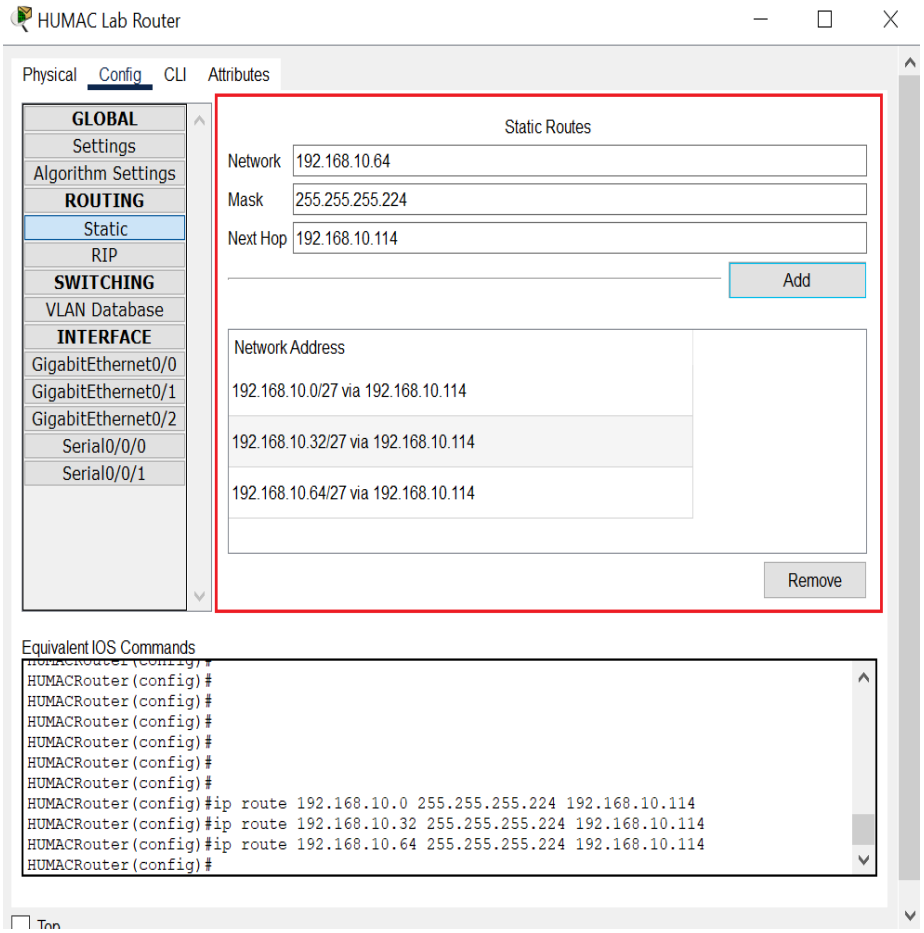


Figure 7.9: Configuring static route of HUMAC router (HUMAC Lab to CL Lab)

Figure 7.9 shows the configuration for the third static route for the HUMAC router. This route is from the HUMAC lab to the CL lab. The IP address 192.168.10.64 is the network address of the CL subnet, and the mask 255.255.255.224 also belongs to the CL subnet. The 'next hop' is the IP address of the Serial0/0/0 interface of the SET router. Now, the HUMAC router is aware of the route to direct messages from the HUMAC lab to the CL lab.

After configuring all the static routes, every device on the network can now communicate with each other despite being in different buildings.

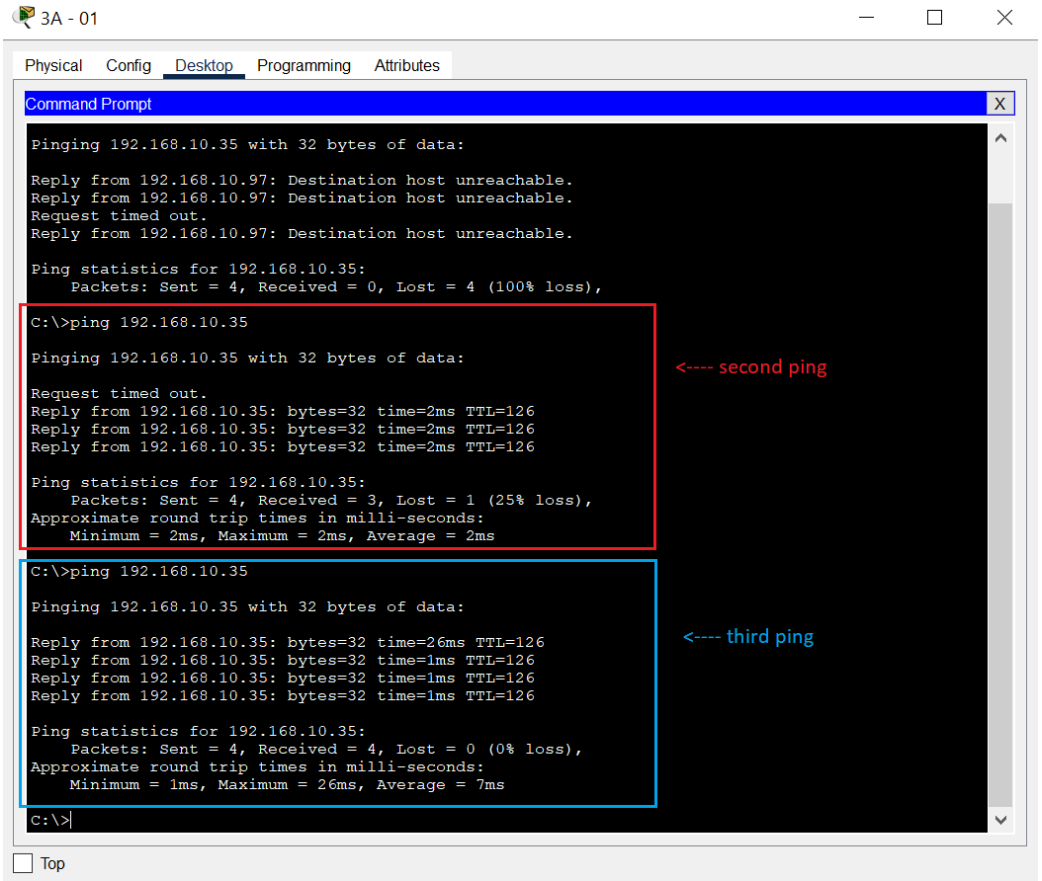


Figure 7.10: Second and Third Ping from a PC in HUMAC lab to a PC in IOT lab

Previously in ***Figure 7.5***, computer 3A – 01 and computer IOT – PC01 from the HUMAC lab and IOT lab respectfully could not connect as all packages sent from 3A-01 to PC01 were lost.

After configuring the static routes for all of the routers, pinging PC01 for the second time with its IP address (192.168.10.35) from 3A-01 will fail because the HUMAC router is not aware of the MAC address of PC01, so an ARP broadcast is sent. Once the MAC address is known and added to the ARP table, the ping will be successful during the third time, meaning that all packets sent and received were successful.

PART C: LESSONS LEARNED

The team collaborated effectively by utilizing everyone's individual strengths and skills to accomplish both design and implementation objectives. Structured task delegation made sure that technical aspects were addressed by the group members with expertise while others concentrated on organization and documentation. Coordination was made easier by frequent check-ins by the leader and open communication, which enabled team members to solve problems together and refine their approach. To avoid bottlenecks and preserve workflow efficiency, soft deadlines were used to guarantee a smooth transition of tasks, from equipment listing to physical and logical topology design. Streamlining task transitions and facilitating the seamless construction of the addressing table and other essential components were made possible by thorough documentation, such as the physical connection table.

During the implementation of the network design (where wireless routers were utilized within the AWL Lab and IOT Lab to enable wireless connectivity with end devices), the team encountered difficulties regarding the connectivity between the AWL Switch and the AWL Wireless Router as well as between the IOT Switch and the IOT Wireless Router. Despite meticulous configuration and verification of IP addresses, subnet masks, and default gateways, there were persistent error messages prompted, preventing the router's WAN interface from establishing a connection. The team's approach to this matter was to collaboratively troubleshoot, with each member attempting various techniques such as ping tests, interface checks and even re-addressing of IP addresses. However, despite these efforts, the underlying cause of the issue was not definitively identified which indicates a possible software malfunction or limitation within the Cisco Packet Tracer.

To effectively resolve this issue, the team utilized online resources such as YouTube tutorials and guides to gain insights and potential solutions. The team also leveraged AI assistance to identify areas that may have been overlooked as well as obtain step-by-step troubleshooting instructions. Failure to recognize the root cause of the issue, despite the aid of digital resources, led us to seek for the lecturer's help, who provided valuable insights and opinions. Ultimately, upon recognizing the probable factor of the persistent errors were the complexity of the routing configuration with the wireless routers, they were replaced with wireless access points. This simplifies the network architecture by operating at Layer 2 and eliminating the need for routing between the devices. Consequently, the conflict was resolved, and the end devices can successfully connect to the network.

Throughout the design and implementation process, the team gained invaluable experience that improved their technical and teamwork skills. They gained a better grasp of network architecture and its real-world applications by going through each stage, from creating equipment inventories to creating physical and logical topologies. Important practical skills like setting up network devices, resolving connectivity problems and recording network topologies for effective task handovers were all reinforced by the project. Additionally, since members needed to effectively cooperate to meet deadlines and overcome obstacles, collaboration, time management and problem-solving skills were crucial takeaways. Ultimately, this practical experience demonstrated the value of learning hands-on in the networking field since it helps people prepare for future professional challenges by bridging the gap between theoritical knowledge and actual application.

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